

EECS 498-008 / 598-008

Deep Learning for Computer Vision

Lecture 1: Introduction

Logistics:

- Remote for first two weeks (Lectures 1 – 3)
- After that, in-person lecture in Chrysler 220

Deep Learning for Computer Vision

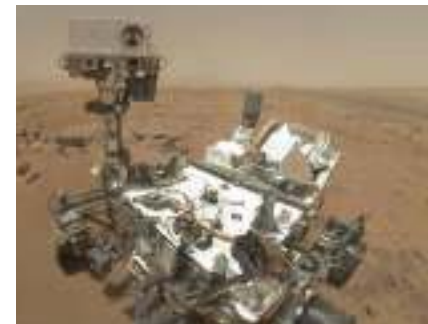
Deep Learning for Computer Vision

Building artificial systems
that process, perceive, and
reason about visual data

Computer Vision is everywhere!



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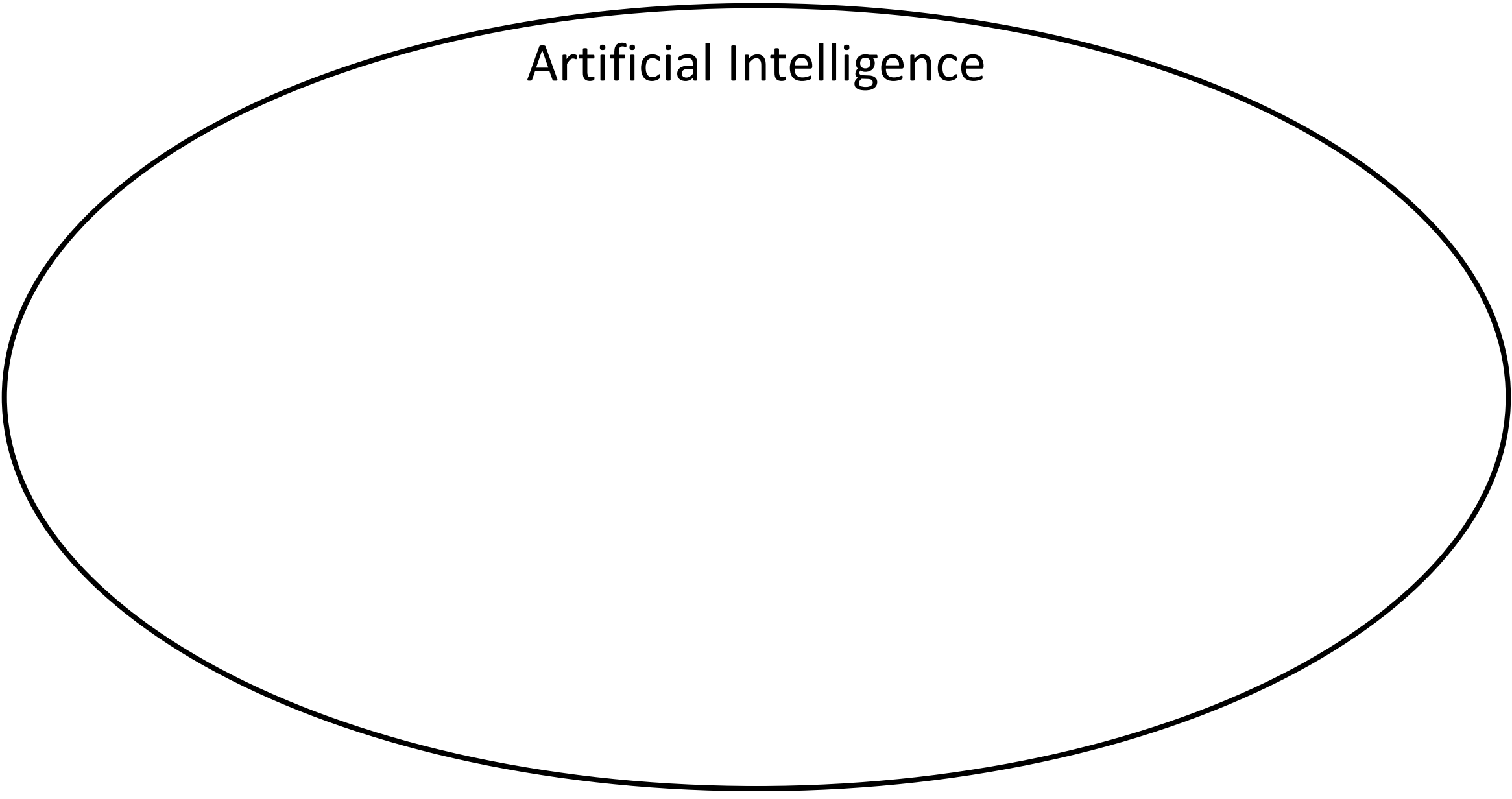
Bottom row, left to right
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Deep Learning for Computer Vision

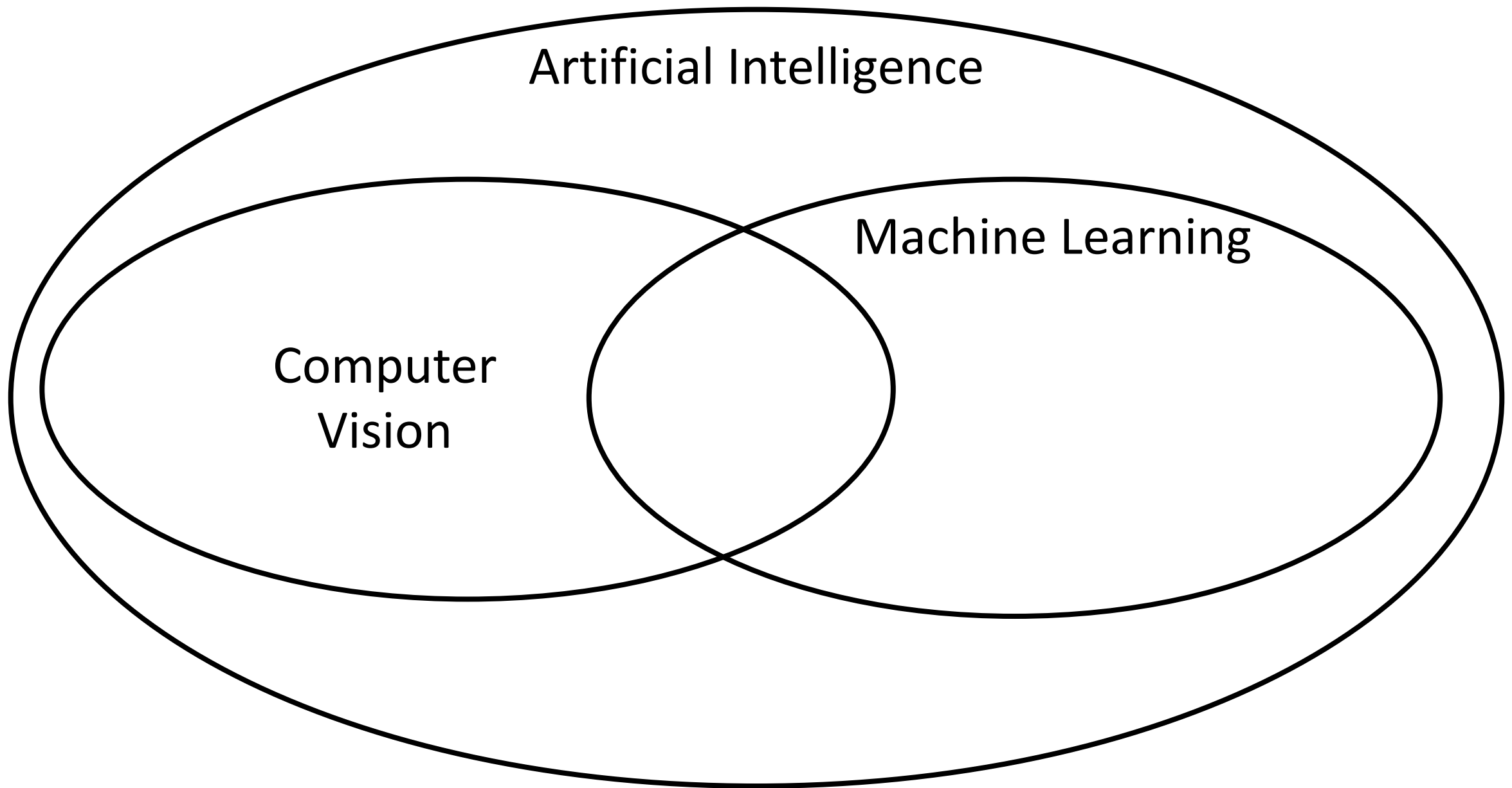
Building artificial systems that
learn from data and experience

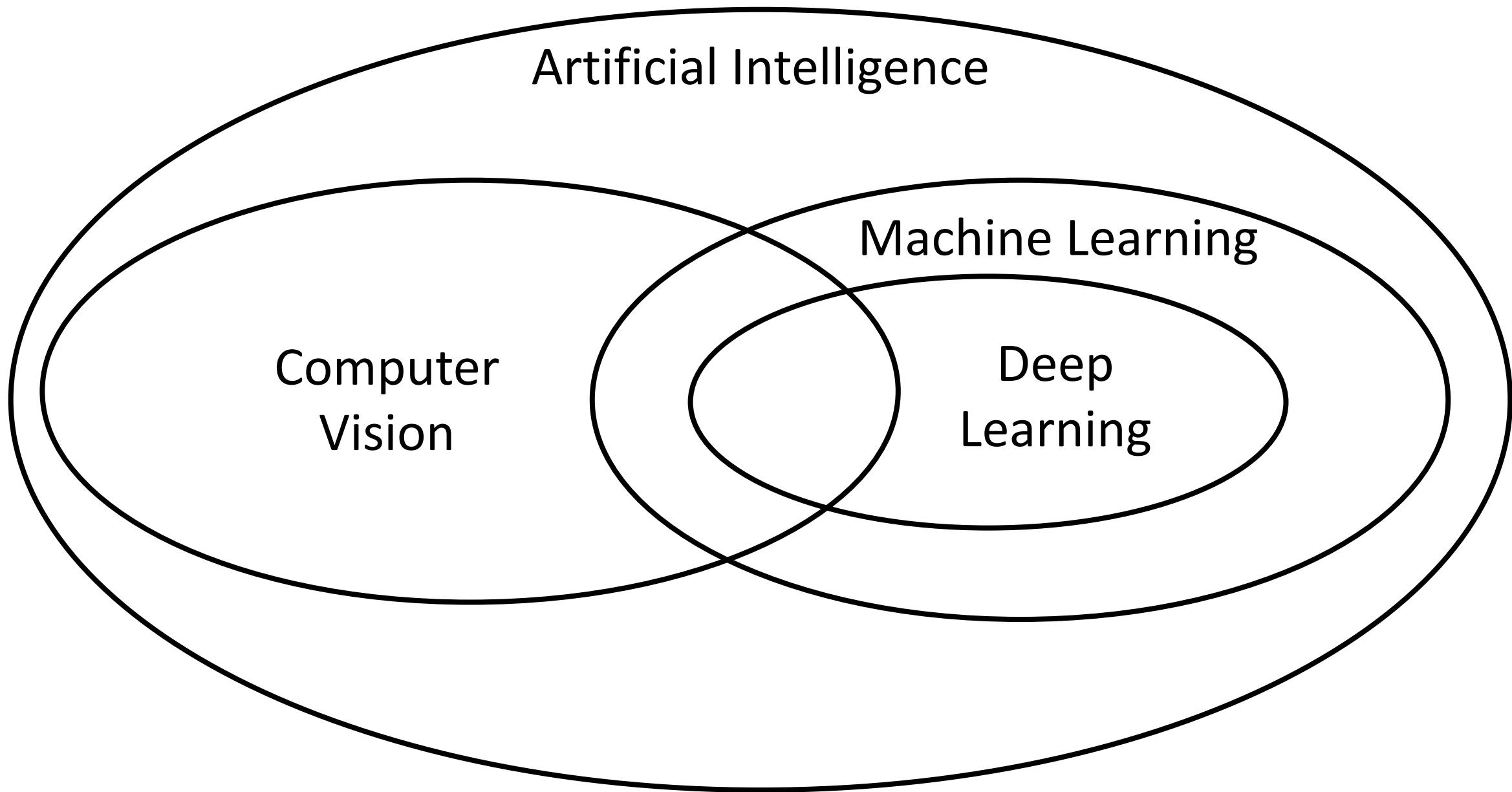
Deep Learning for Computer Vision

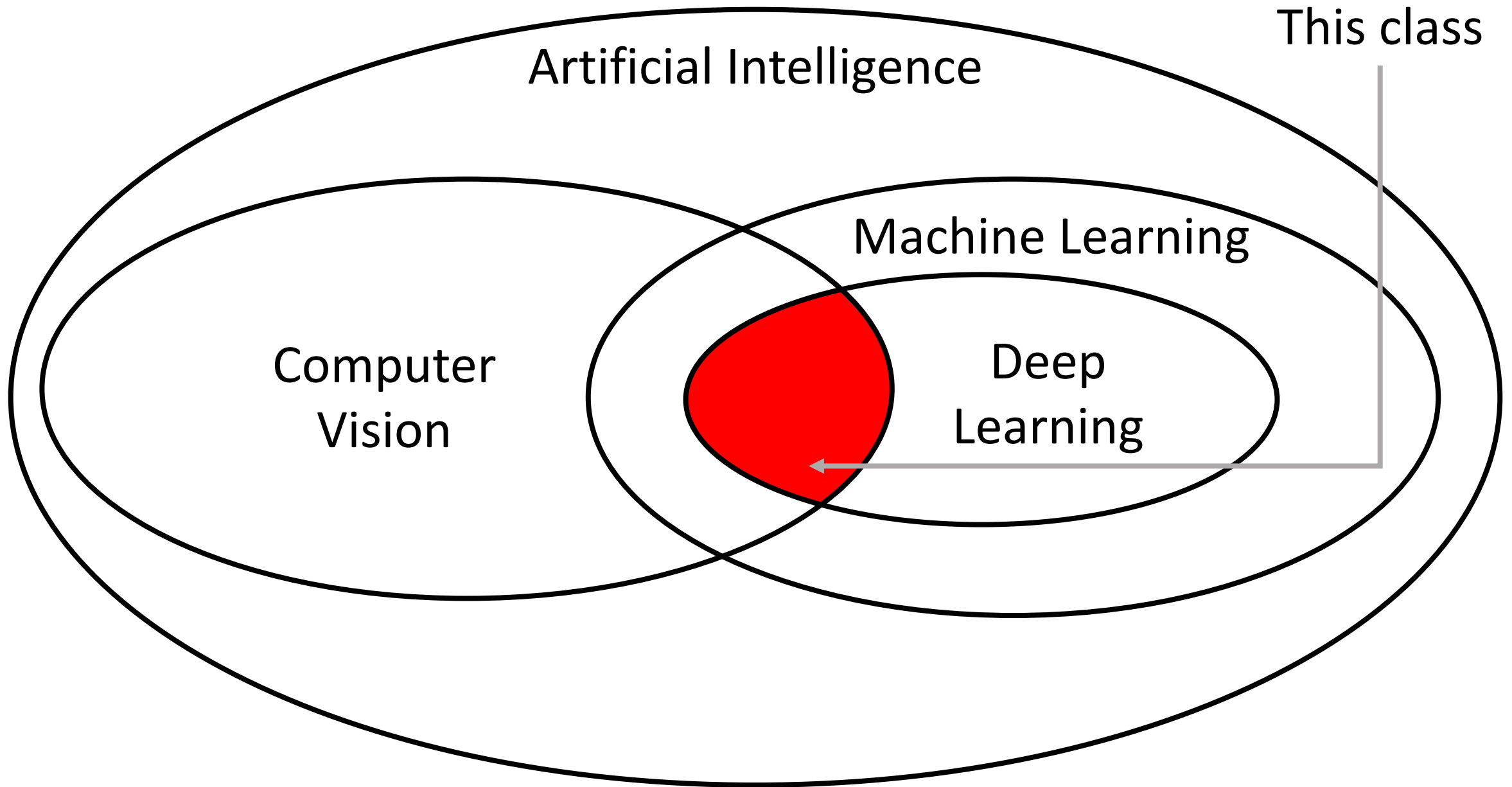
Hierarchical learning algorithms
with many “layers”, (very) loosely
inspired by the brain

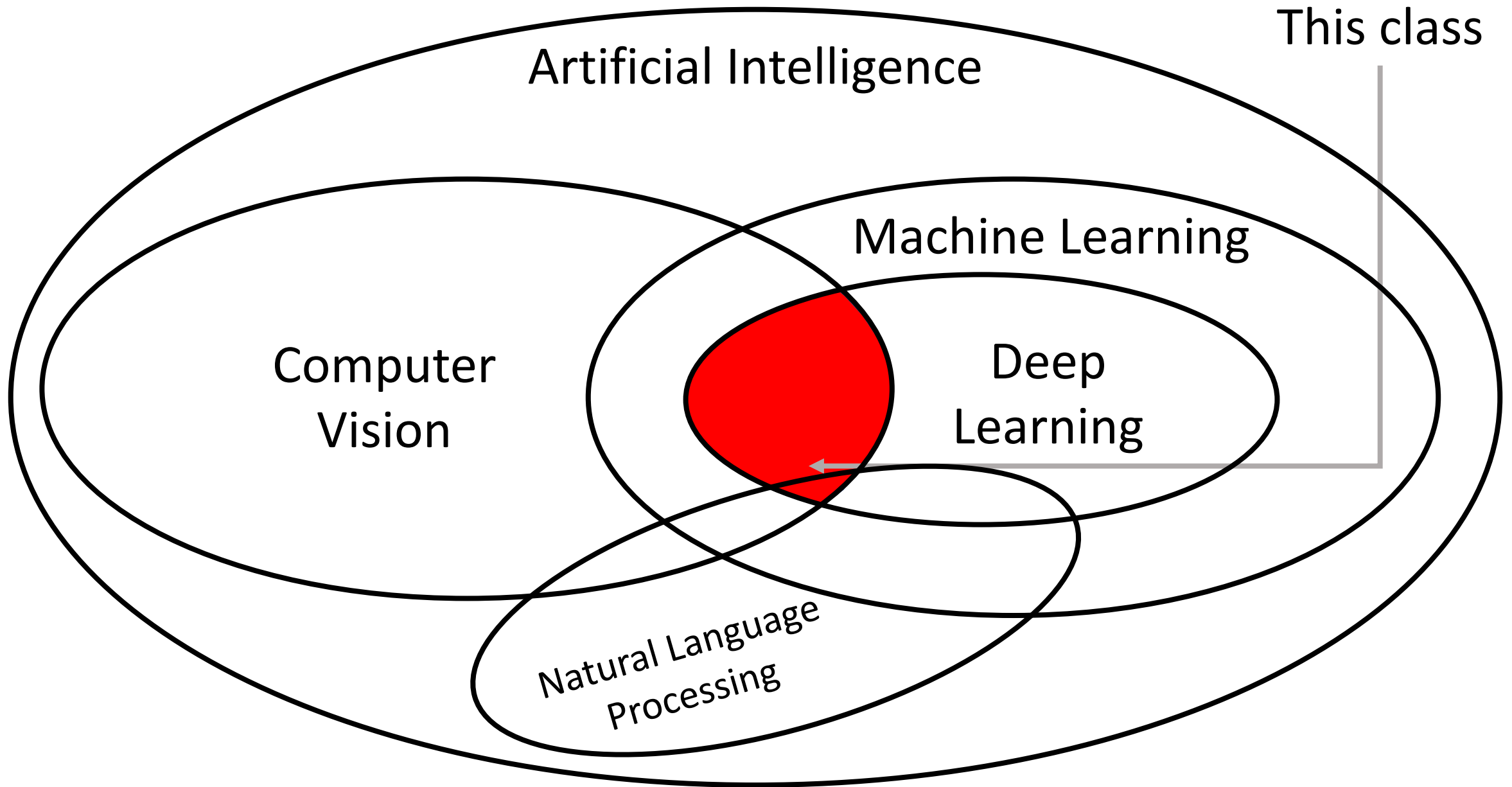


Artificial Intelligence









This class

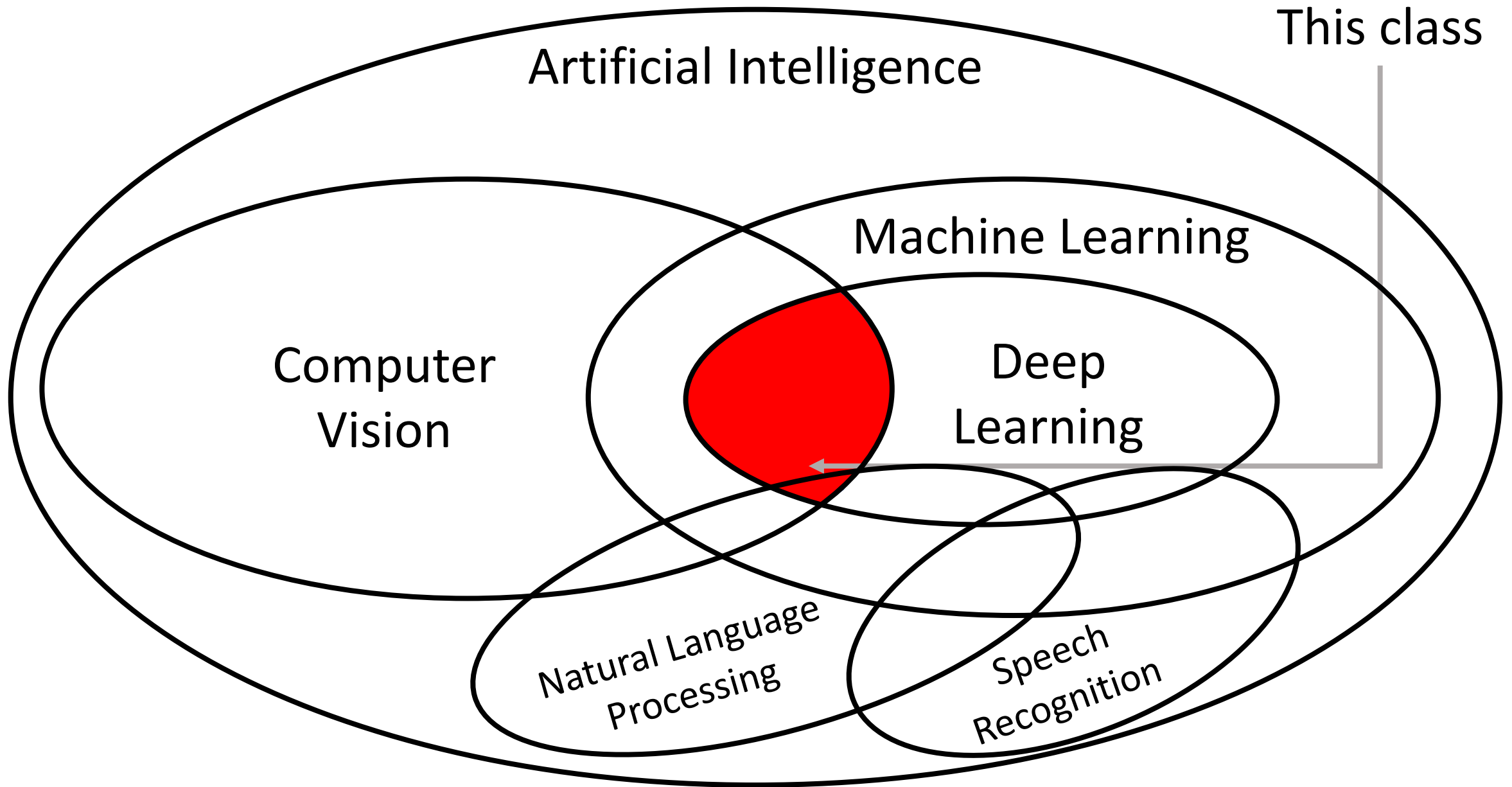
Artificial Intelligence

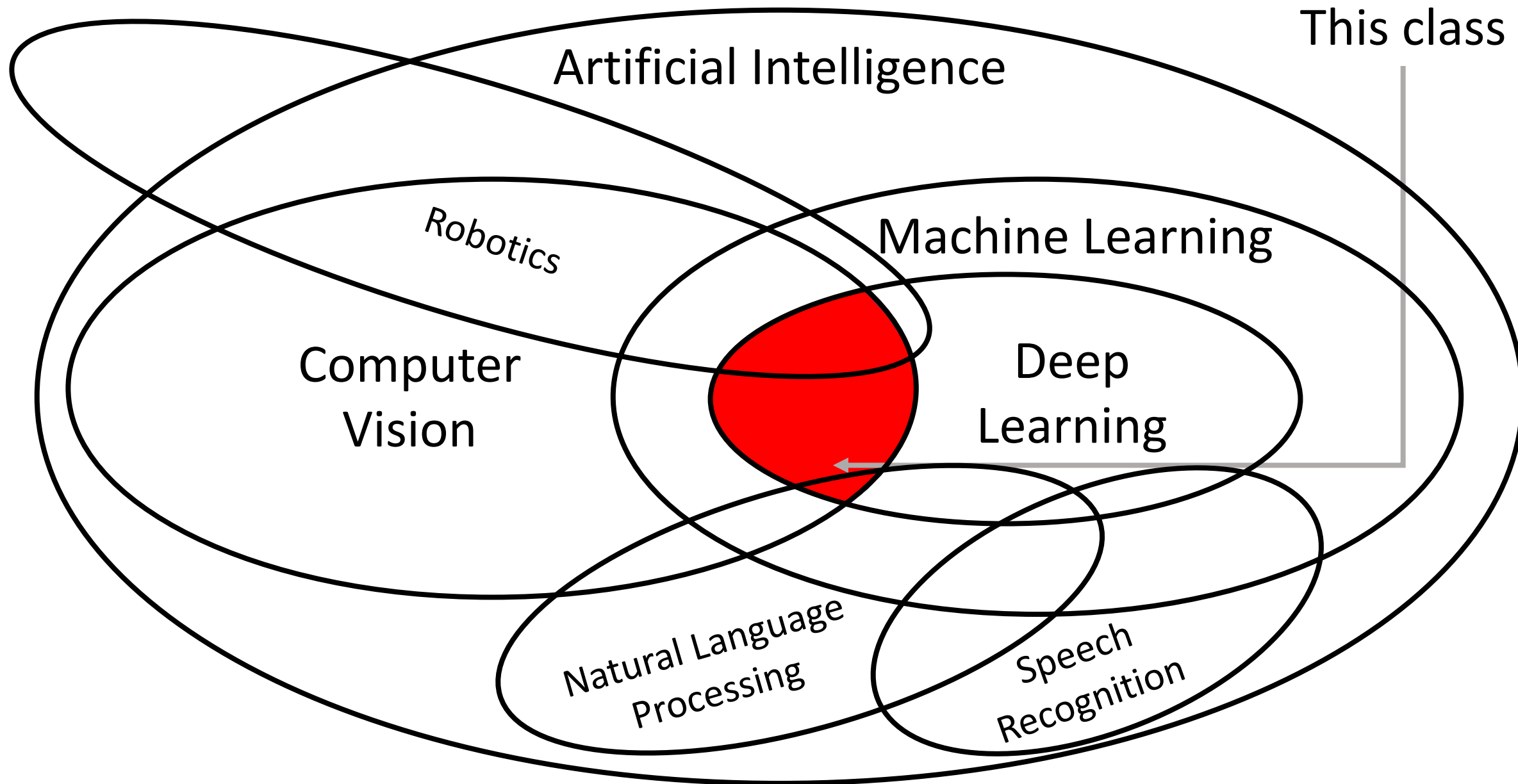
Machine Learning

Computer
Vision

Deep
Learning

Natural Language
Processing





This class

Artificial Intelligence

Machine Learning

Deep Learning

Computer Vision

Robotics

Natural Language Processing

Speech Recognition

Today's Agenda

- A brief history of computer vision and deep learning
- Course overview and logistics

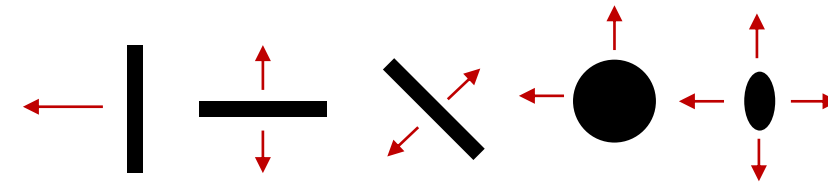
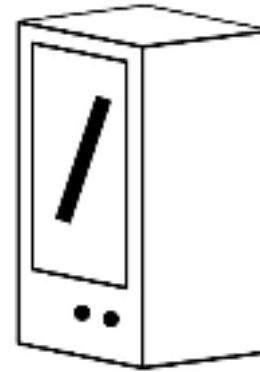
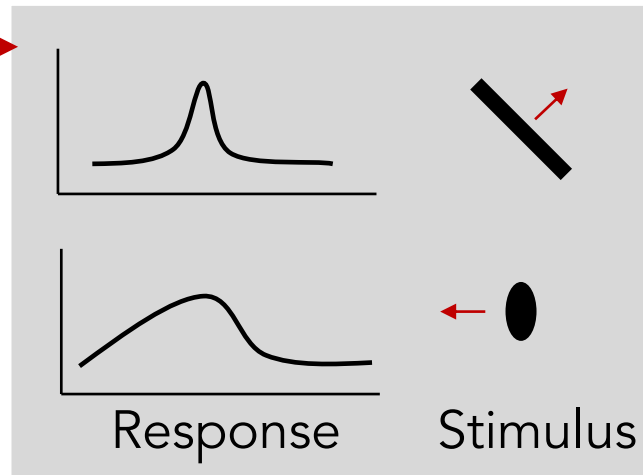
Hubel and Wiesel, 1959

Measure
brain activity

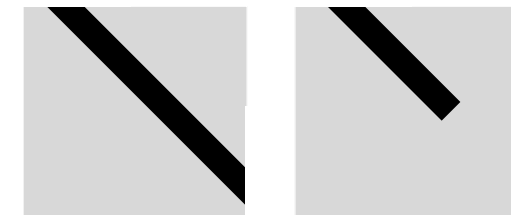


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1959
Hubel & Wiesel



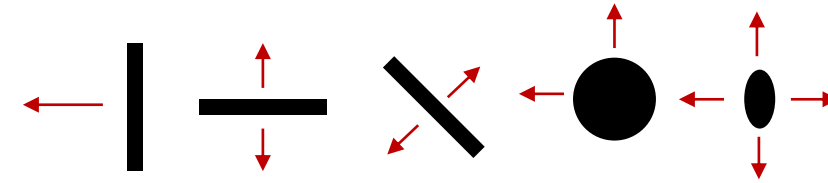
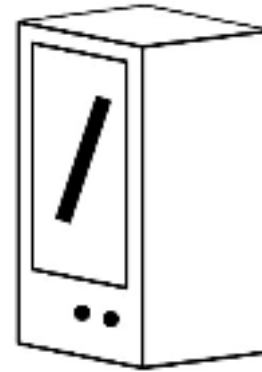
Simple cells:
Response to light
orientation



No response

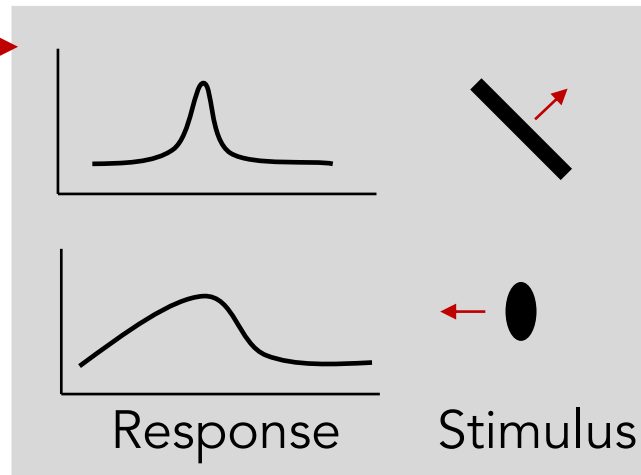
Hubel and Wiesel, 1959

Measure
brain activity



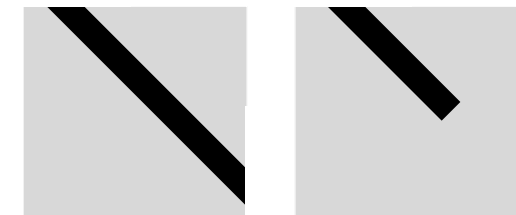
Simple cells:
Response to light
orientation

Complex cells:
Response to light
orientation and
movement



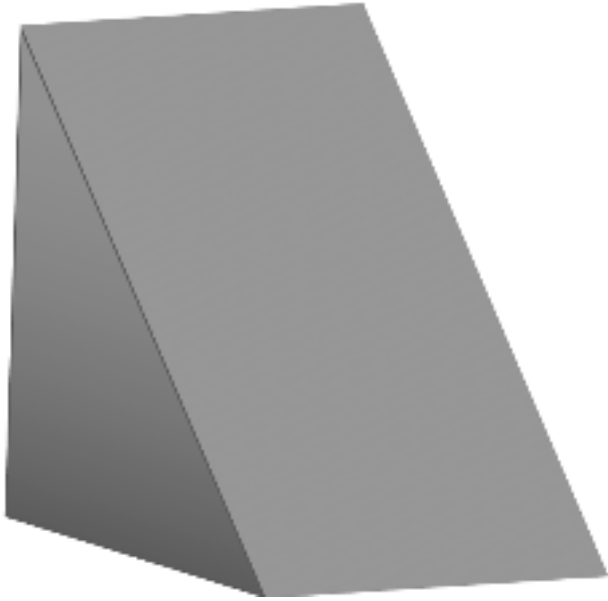
Cat image by CNX OpenStax is licensed
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1959
Hubel & Wiesel

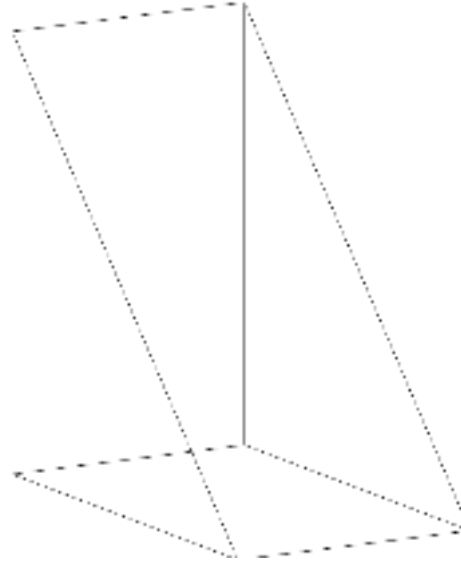


No response

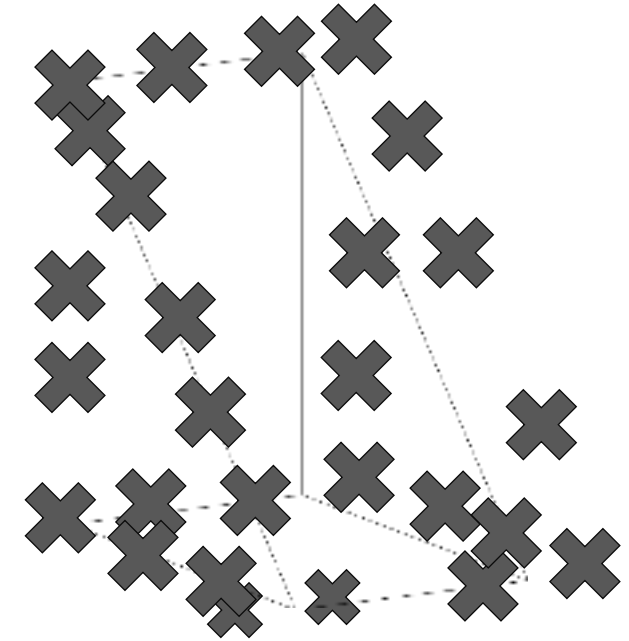
Larry Roberts, 1963



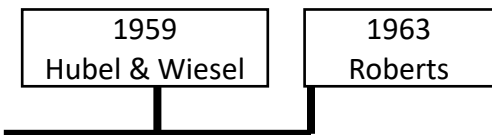
(a) Original picture



(b) Differentiated picture



(c) Feature points selected



Lawrence Gilman Roberts, "Machine Perception of Three-Dimensional Solids", 1963

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

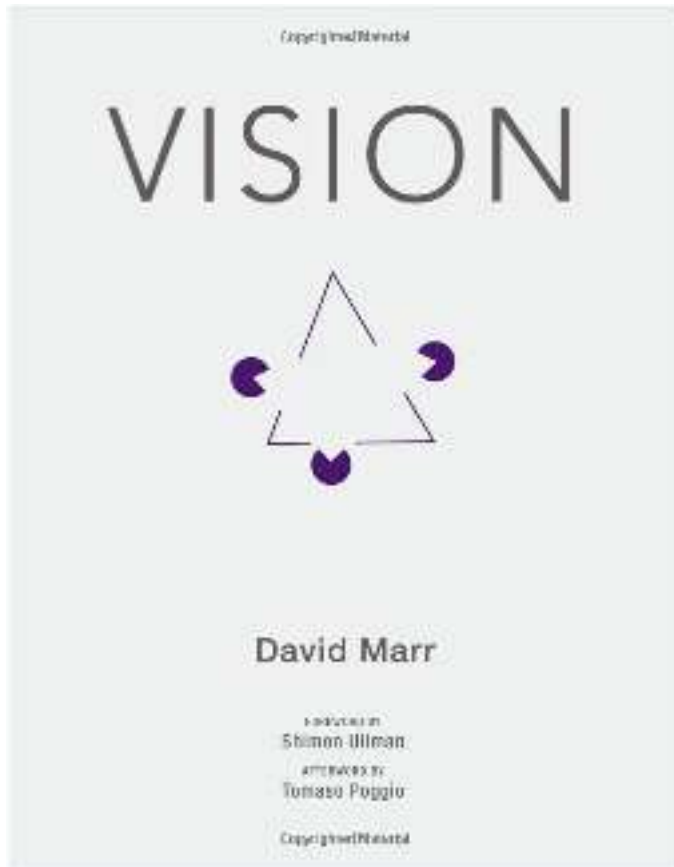
Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

1959
Hubel & Wiesel

1963
Roberts

<https://dspace.mit.edu/handle/1721.1/6125>

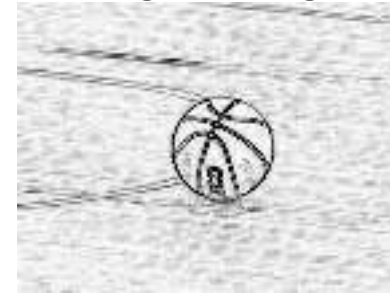


Input image

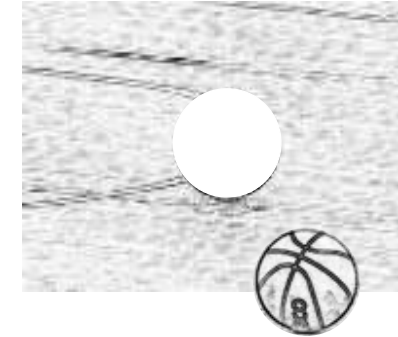


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Edge image



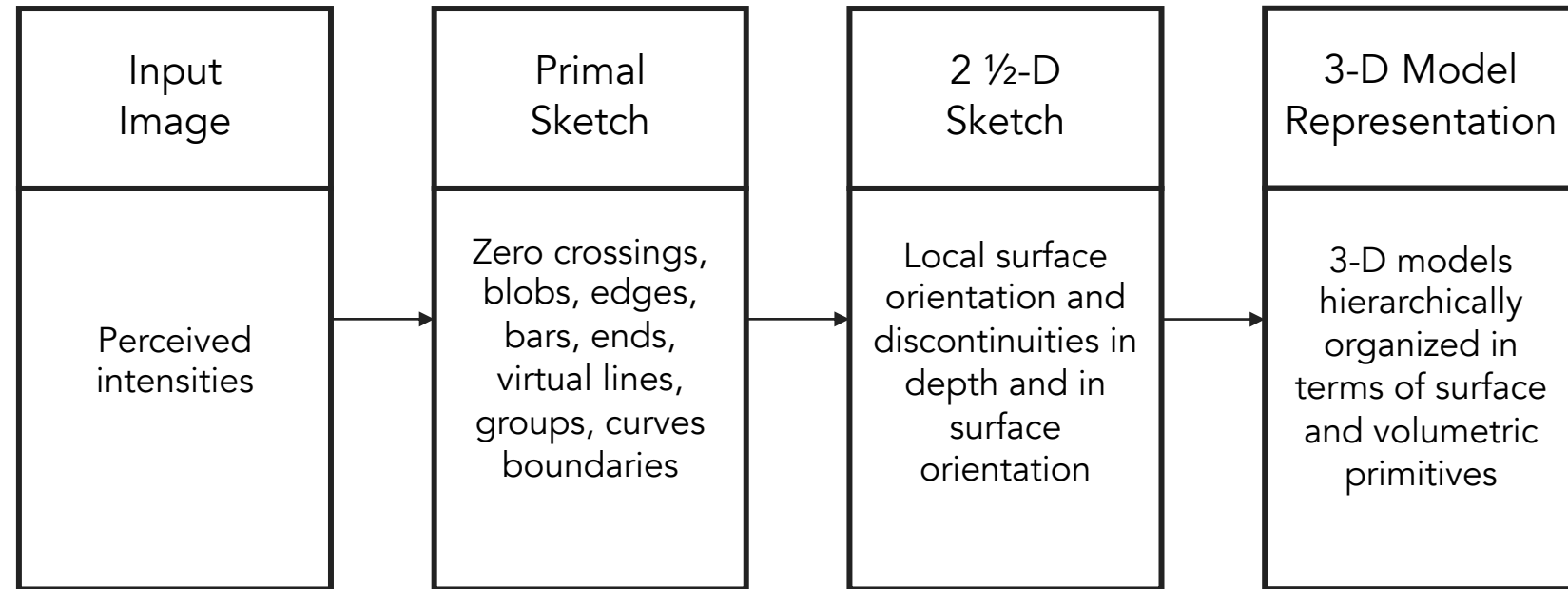
2 1/2-D sketch



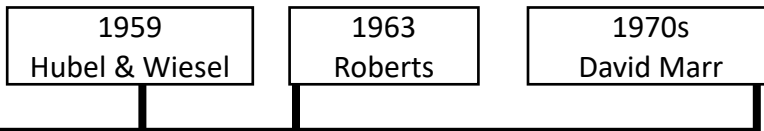
3-D model



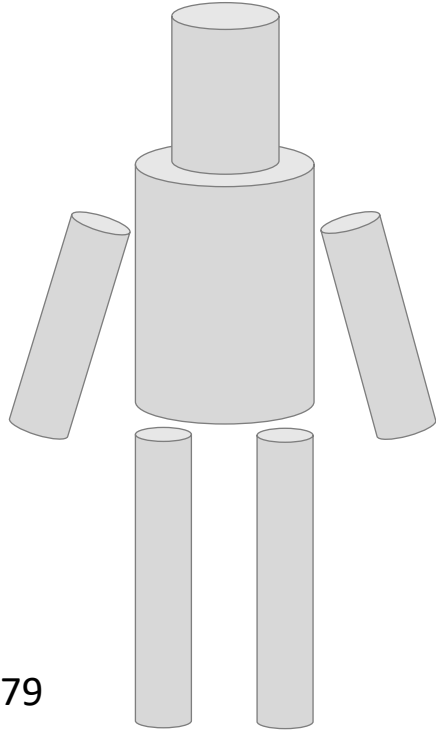
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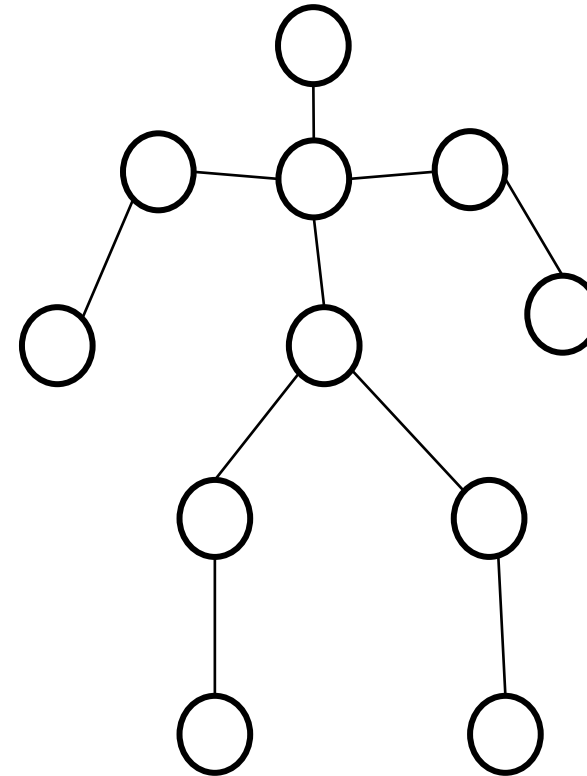
Stages of Visual Representation, David Marr, 1970s



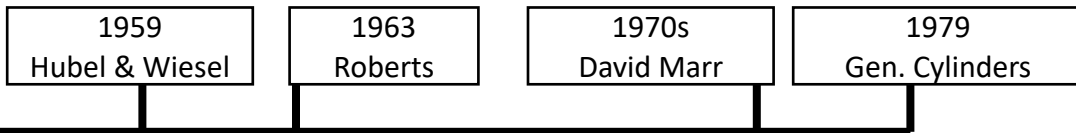
Recognition via Parts (1970s)



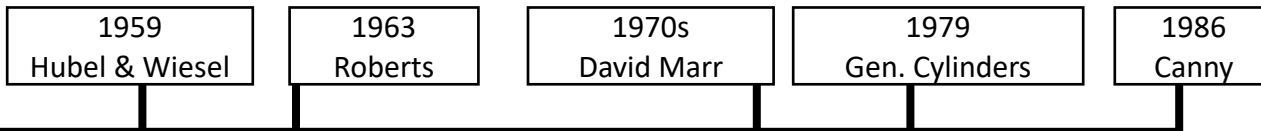
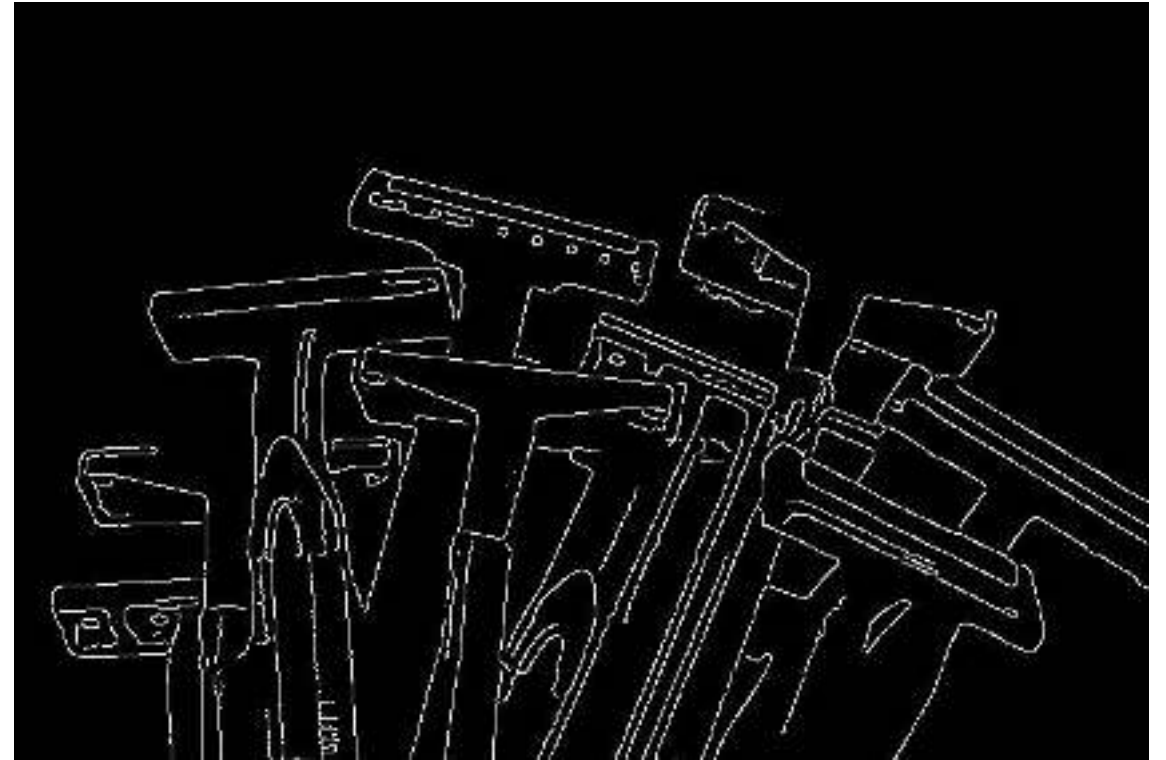
Generalized Cylinders,
Brooks and Binford, 1979



Pictorial Structures,
Fischler and Elshlager, 1973



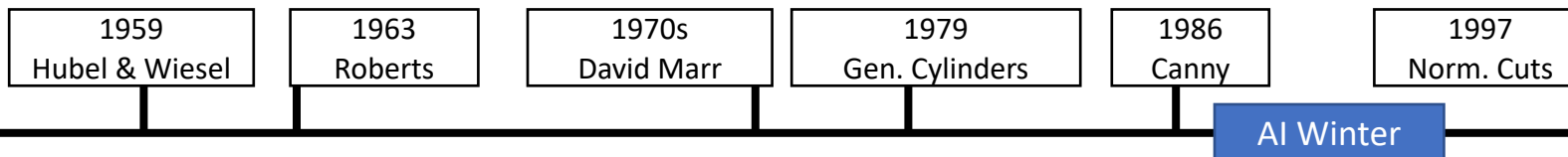
Recognition via Edge Detection (1980s)



John Canny, 1986
David Lowe, 1987

Image is CC0 1.0 public domain

Recognition via Grouping (1990s)



Normalized Cuts, Shi and Malik, 1997

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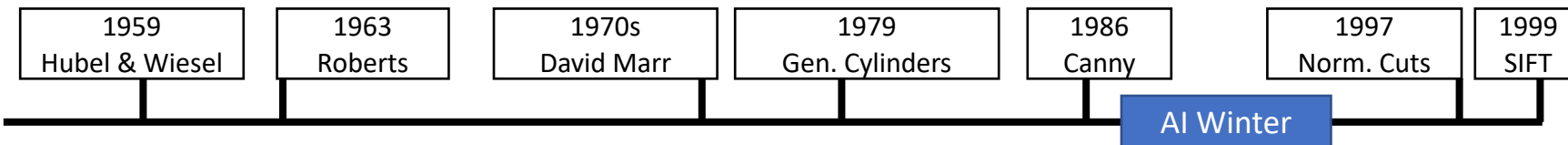
Recognition via Matching (2000s)



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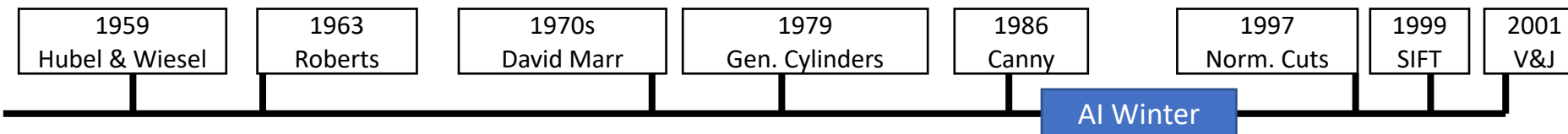


SIFT, David Lowe, 1999

Face Detection

Viola and Jones, 2001

One of the first successful applications of machine learning to vision

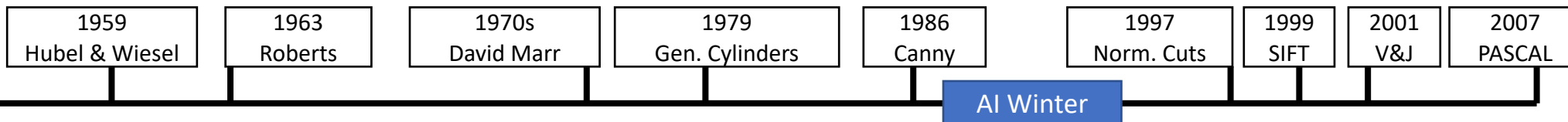
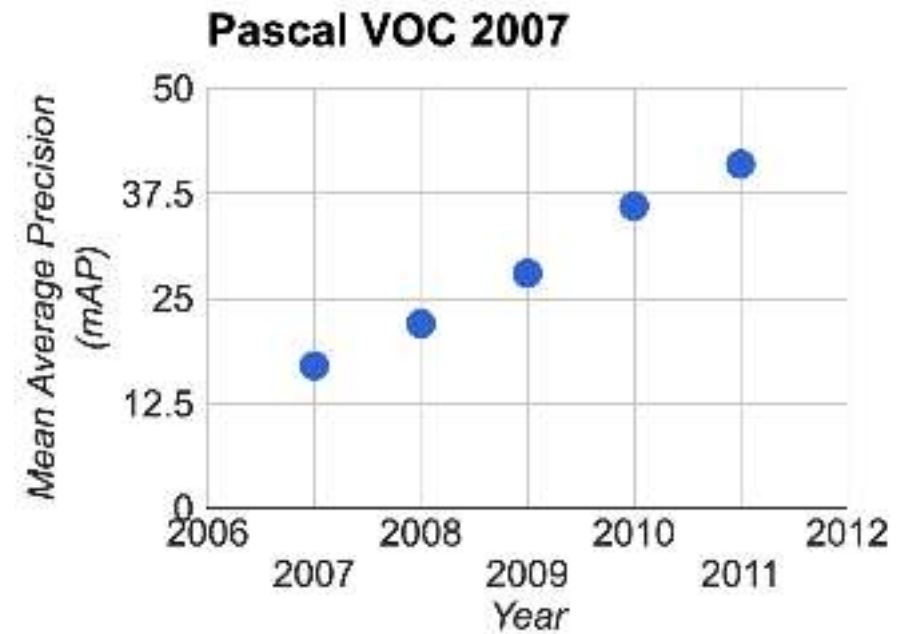


PASCAL Visual Object Challenge

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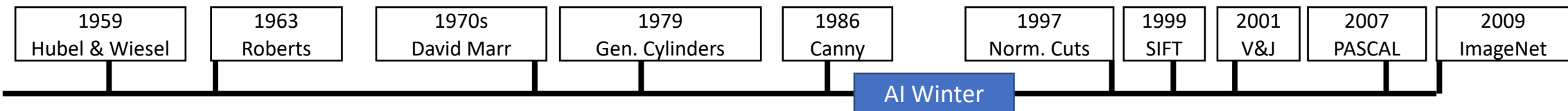
IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:
1,000 object classes
1,431,167 images

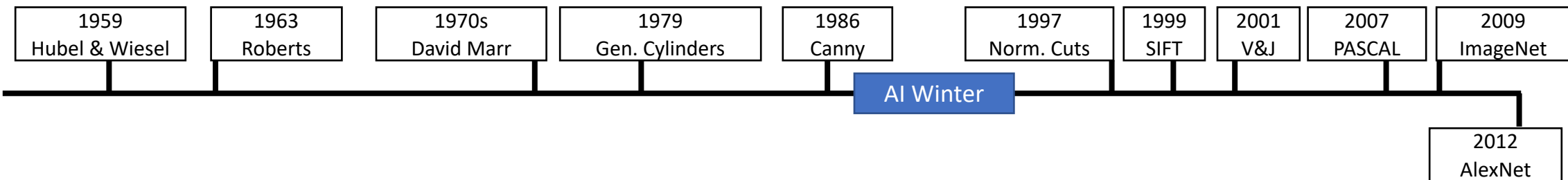
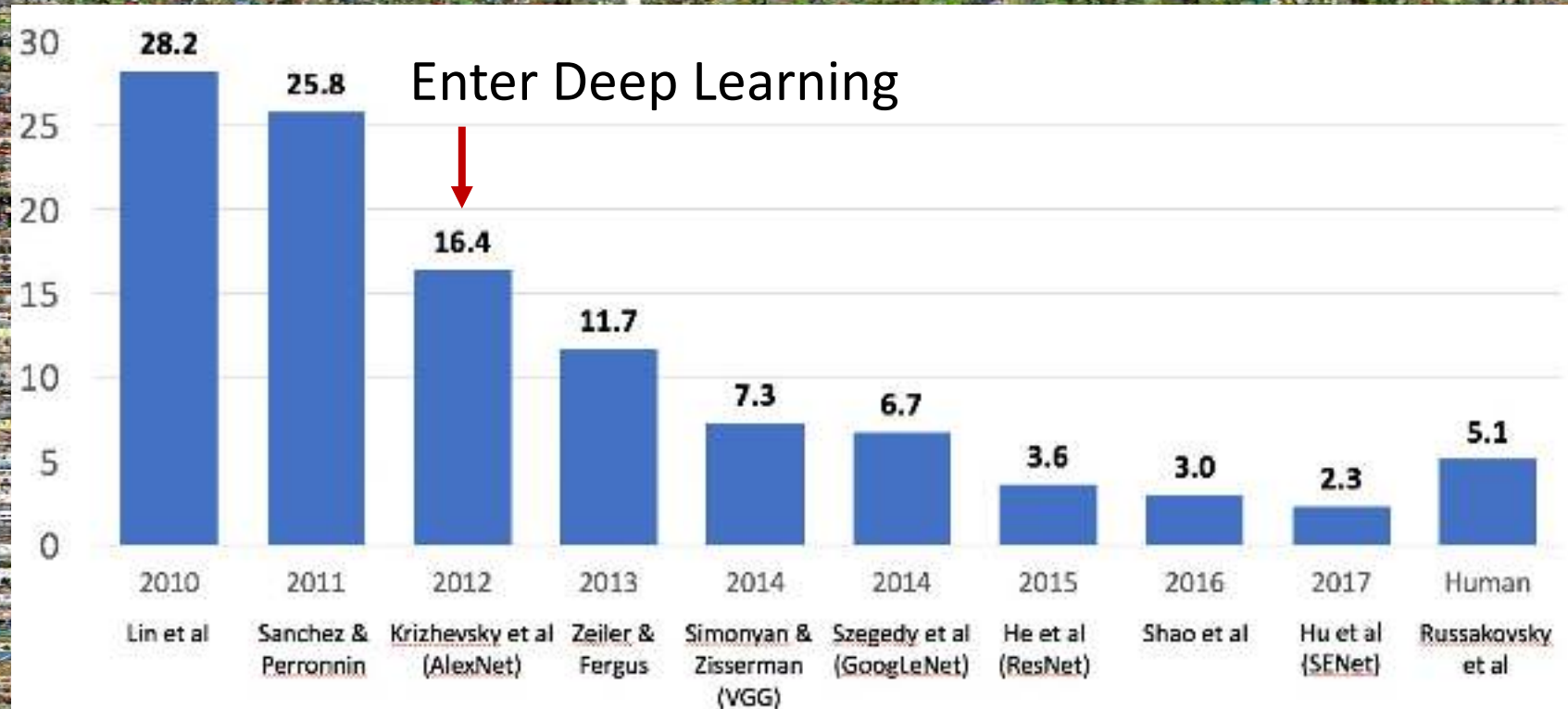


Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle

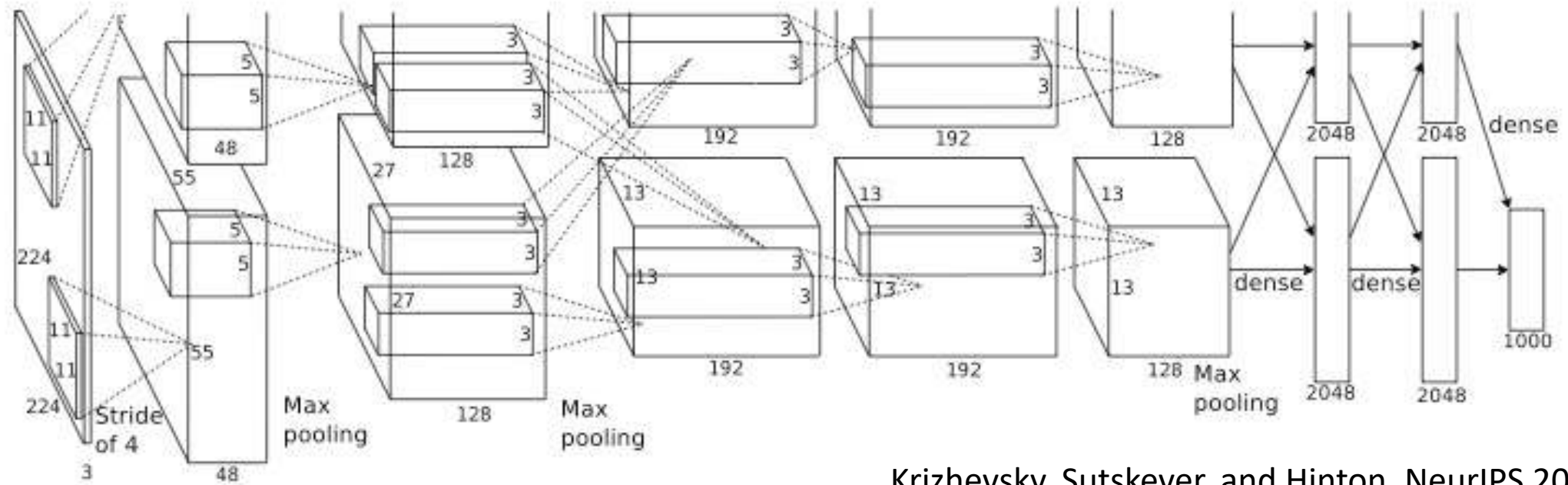
Deng et al, 2009
Russakovsky et al. IJCV 2015



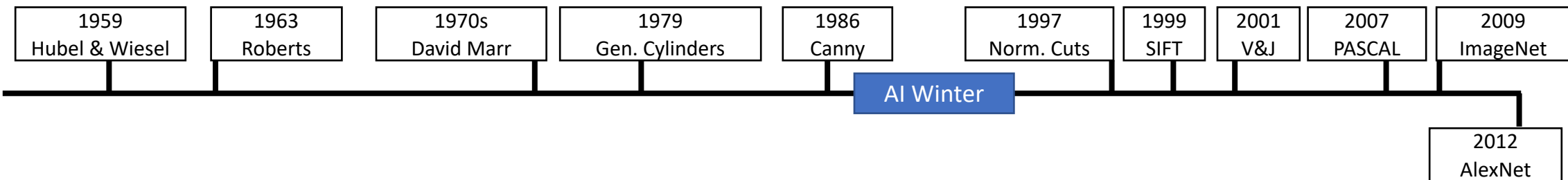
IMAGENET Large Scale Visual Recognition Challenge



AlexNet: Deep Learning Goes Mainstream



Krizhevsky, Sutskever, and Hinton, NeurIPS 2012



Perceptron

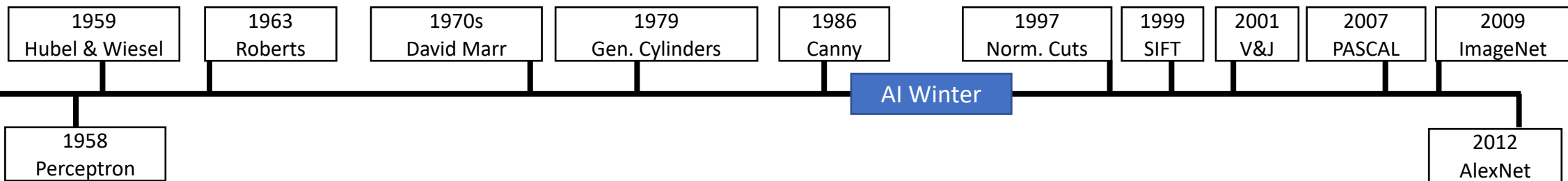
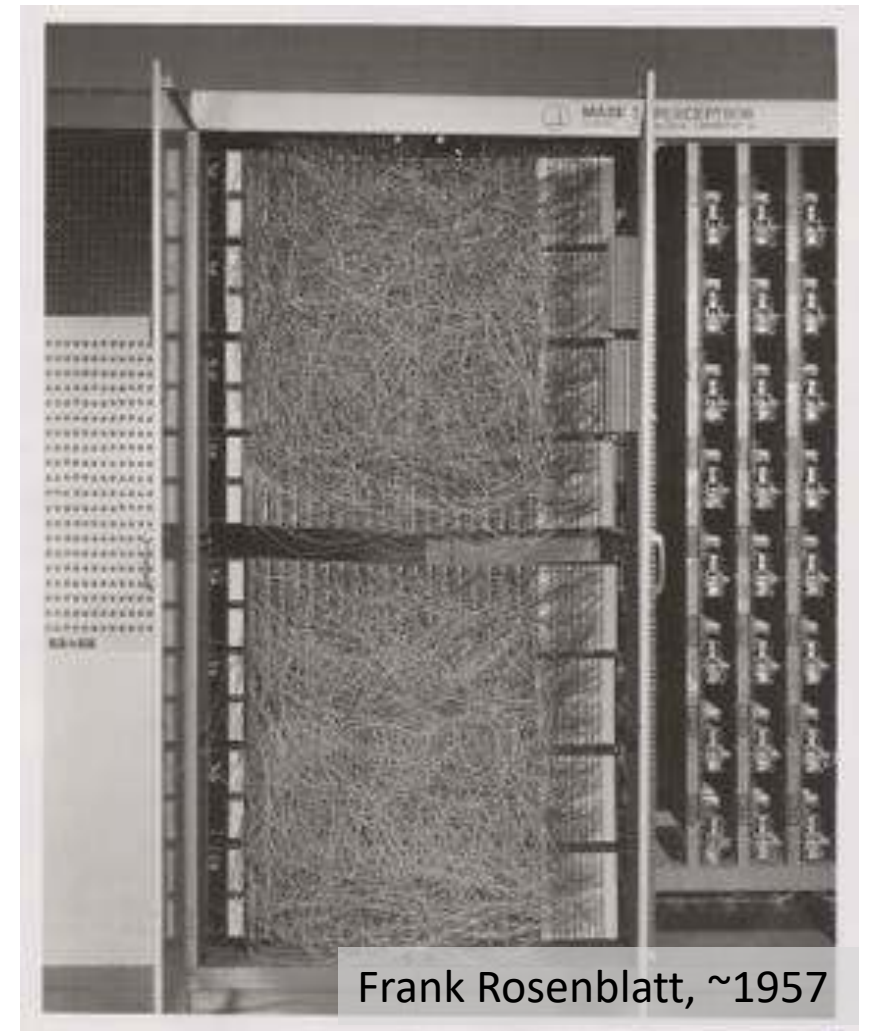
One of the earliest algorithms that could learn from data

Implemented in hardware! Weights stored in potentiometers, updated with electric motors during learning

Connected to a camera that used 20x20 cadmium sulfide photocells to make a 400-pixel image

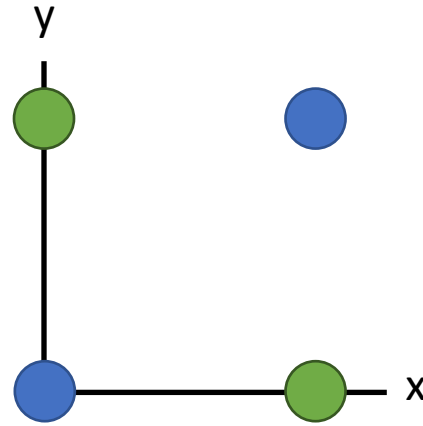
Could learn to recognize letters of the alphabet

Today we would recognize it as a **linear classifier**

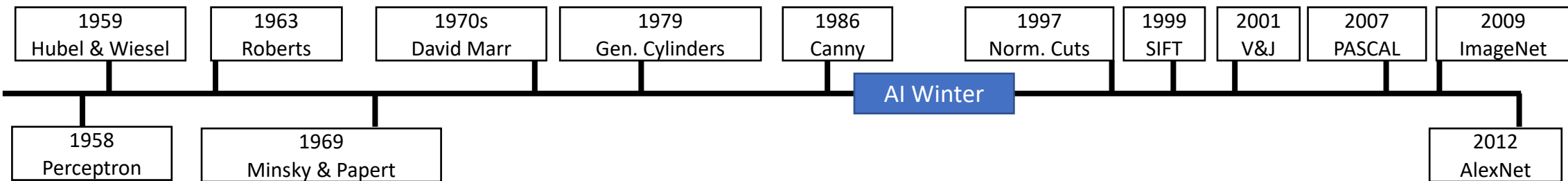
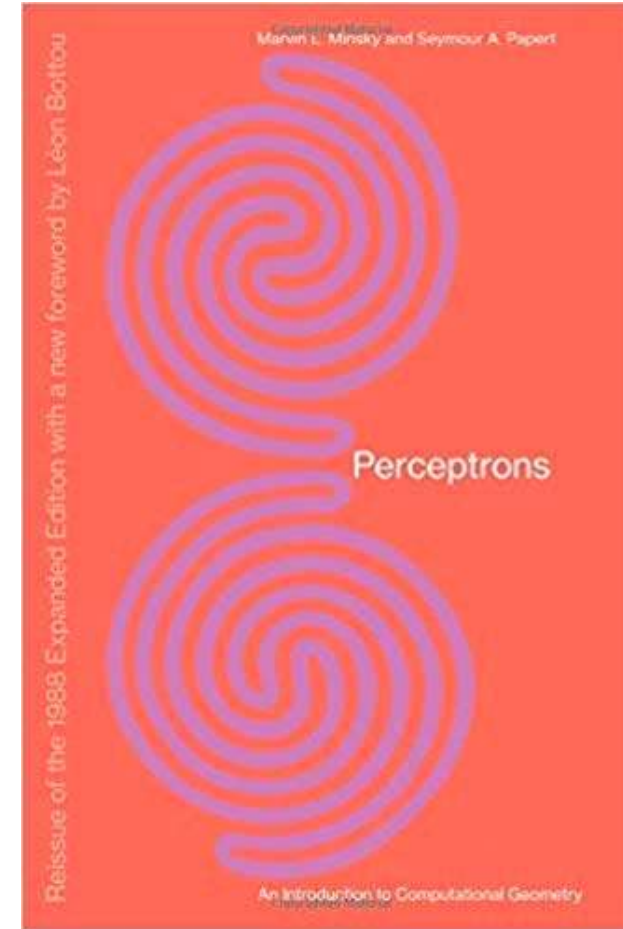


Minsky and Papert, 1969

X	Y	F(x,y)
0	0	0
0	1	1
1	0	1
1	1	0



Showed that Perceptrons could not learn the XOR function
Caused a lot of disillusionment in the field

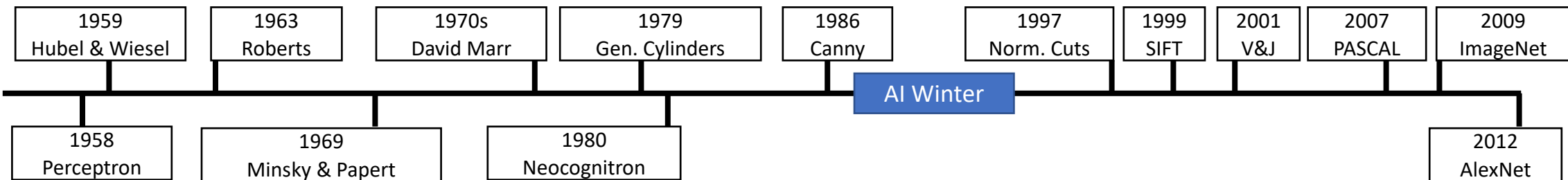
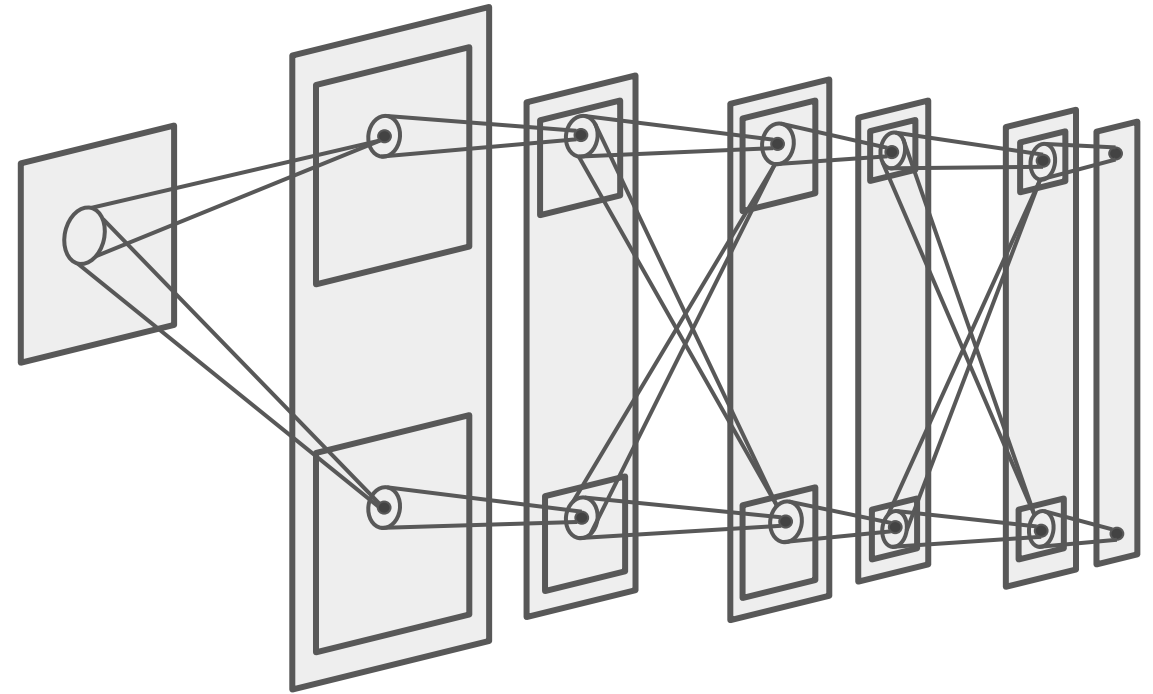


Neocognitron: Fukushima, 1980

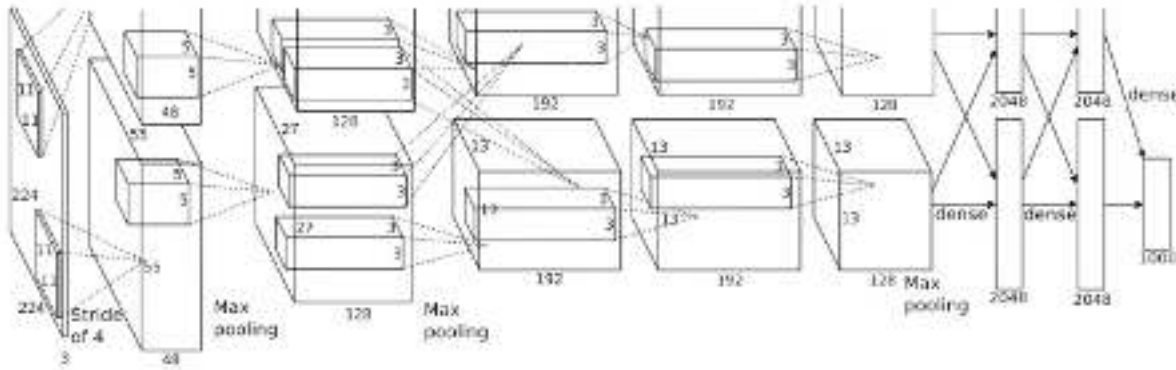
Computational model the visual system,
directly inspired by Hubel and Wiesel's
hierarchy of complex and simple cells

Interleaved simple cells (convolution)
and complex cells (pooling)

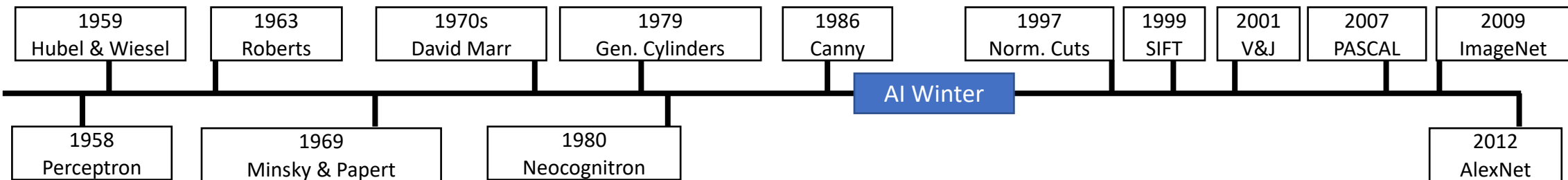
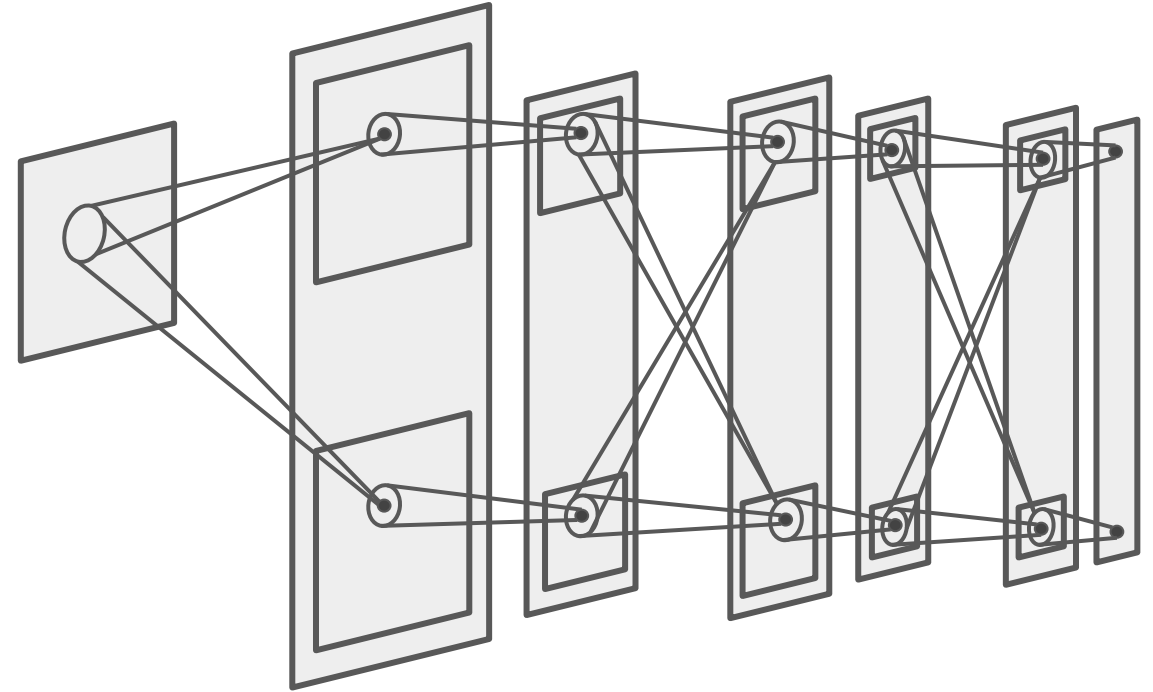
No practical training algorithm



Neocognitron: Fukushima, 1980



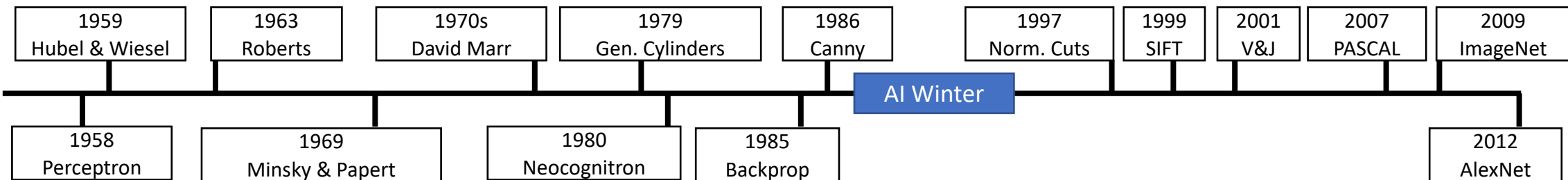
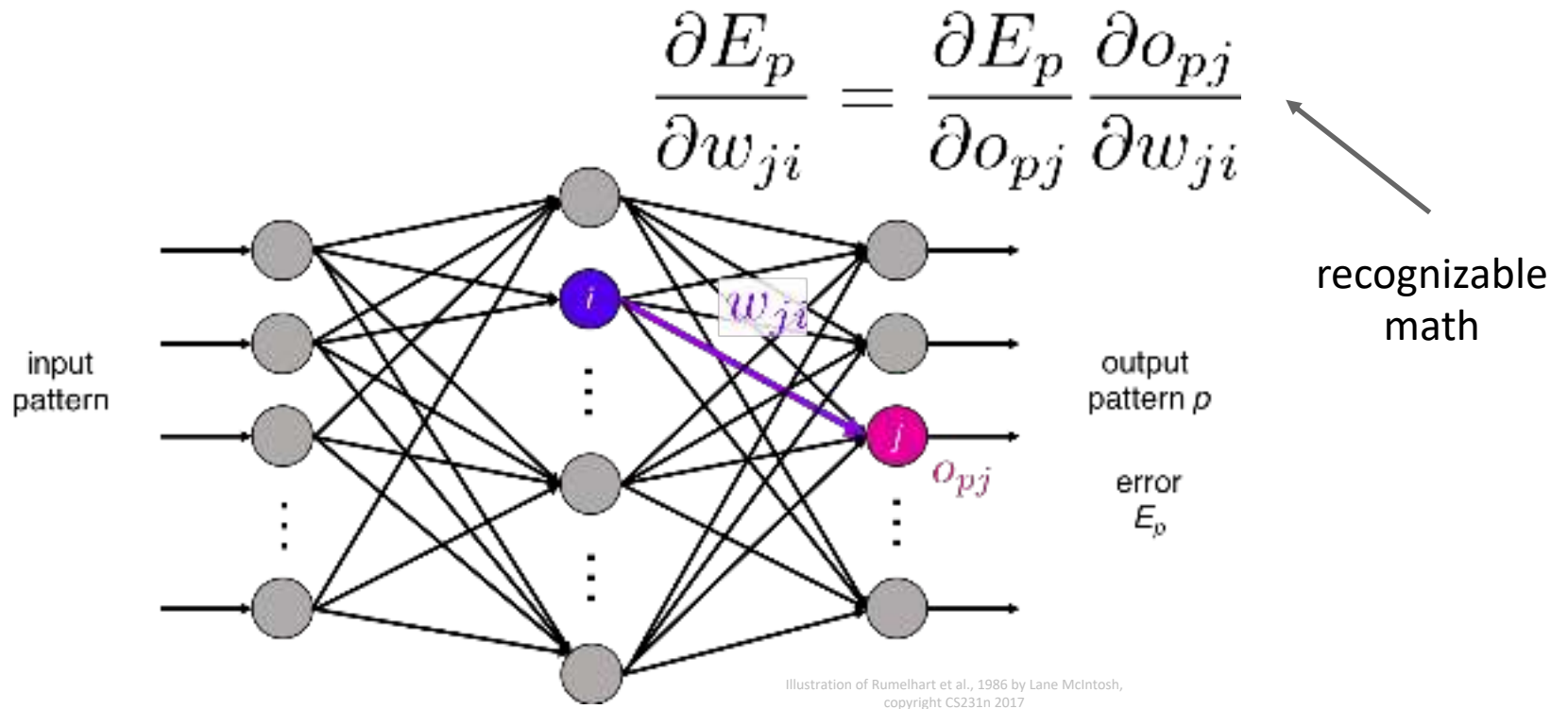
Looks a lot like AlexNet
more than 32 years later!



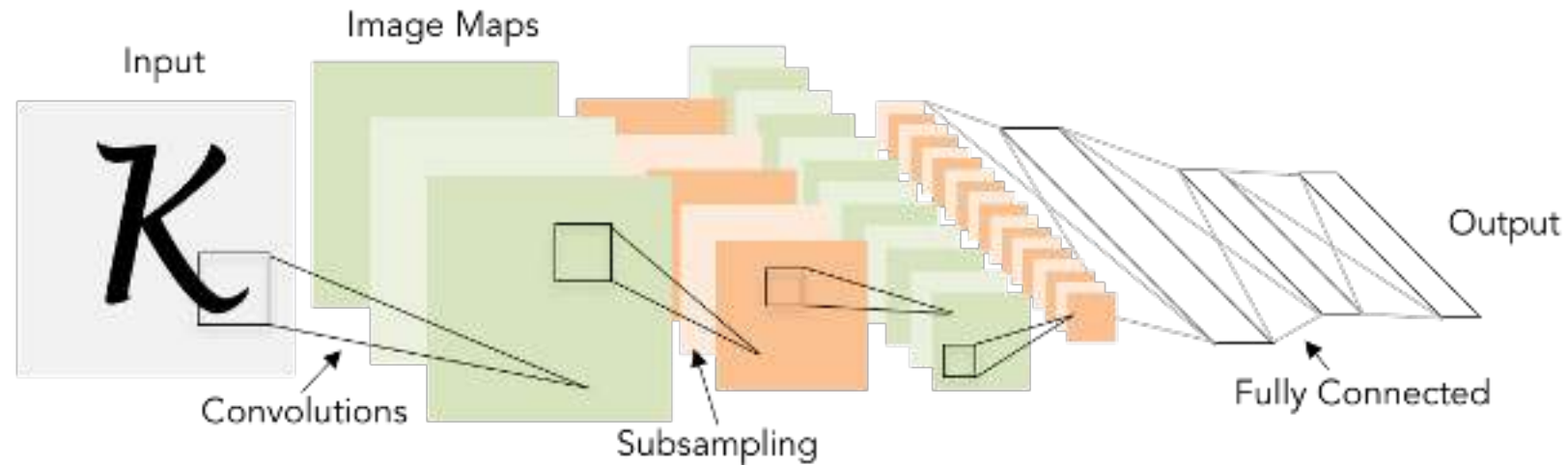
Backprop: Rumelhart, Hinton, and Williams, 1986

Introduced
backpropagation for
computing gradients
in neural networks

Successfully trained
perceptrons with
multiple layers



Convolutional Networks: LeCun et al, 1998

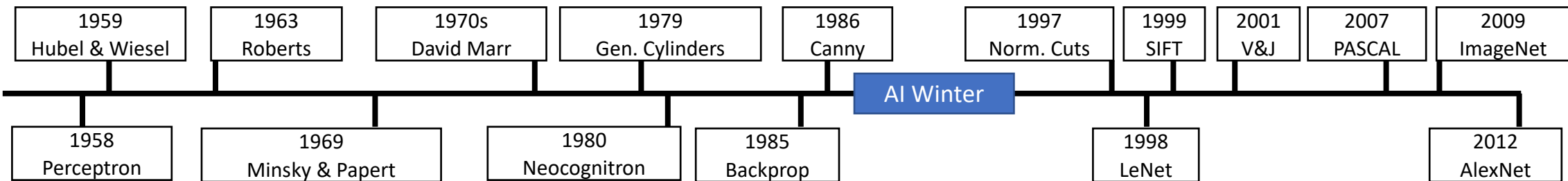


Applied backprop algorithm to a Neocognitron-like architecture

Learned to recognize handwritten digits

Was deployed in a commercial system by NEC, processed handwritten checks

Very similar to our modern convolutional networks!



2000s: “Deep Learning”

People tried to train neural networks that were deeper and deeper

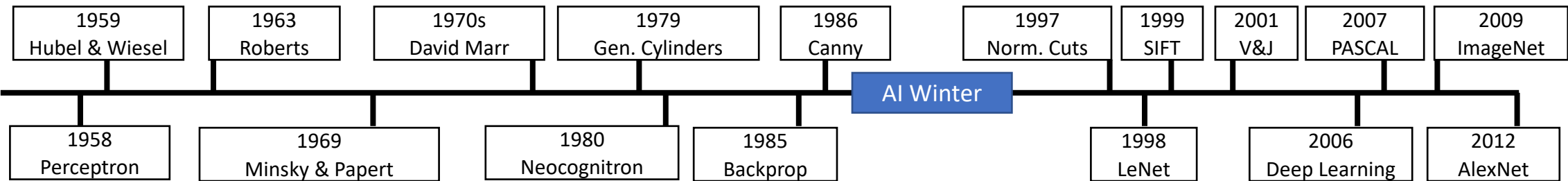
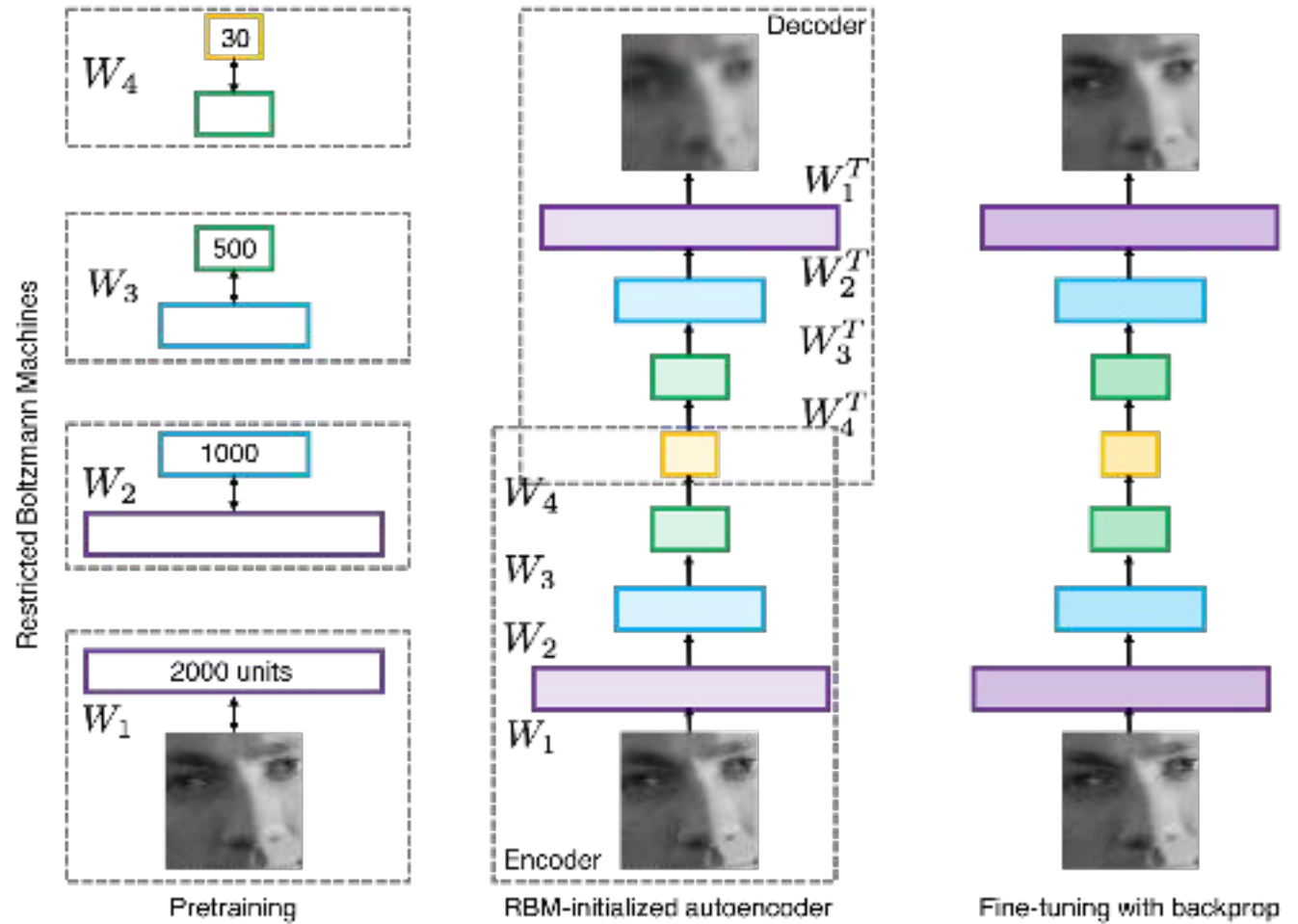
Not a mainstream research topic at this time

Hinton and Salakhutdinov, 2006

Bengio et al, 2007

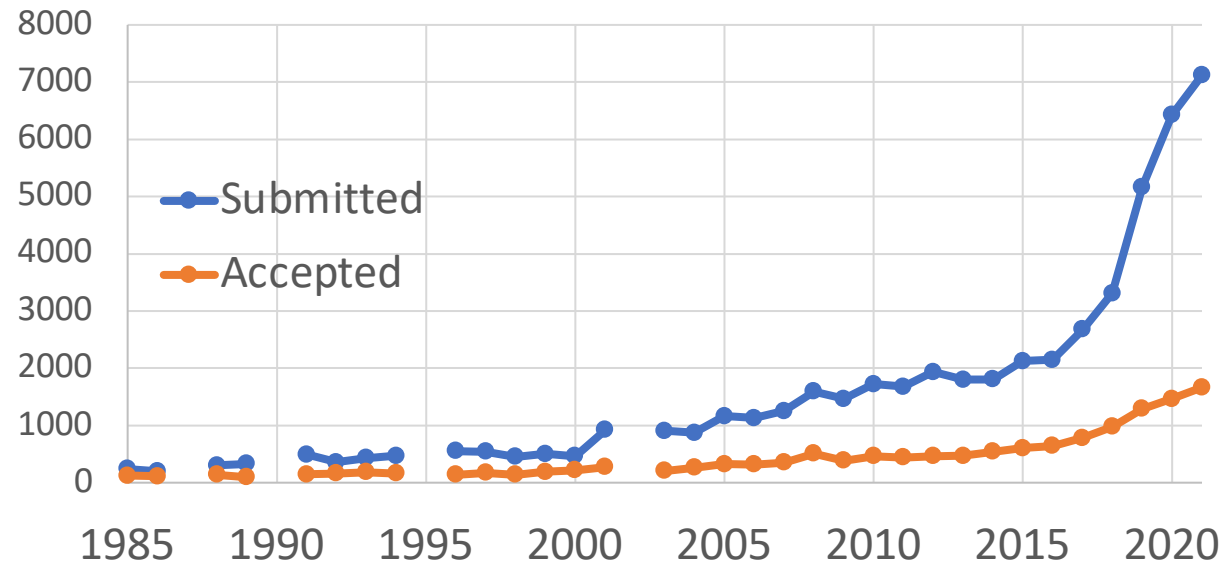
Lee et al, 2009

Glorot and Bengio, 2010

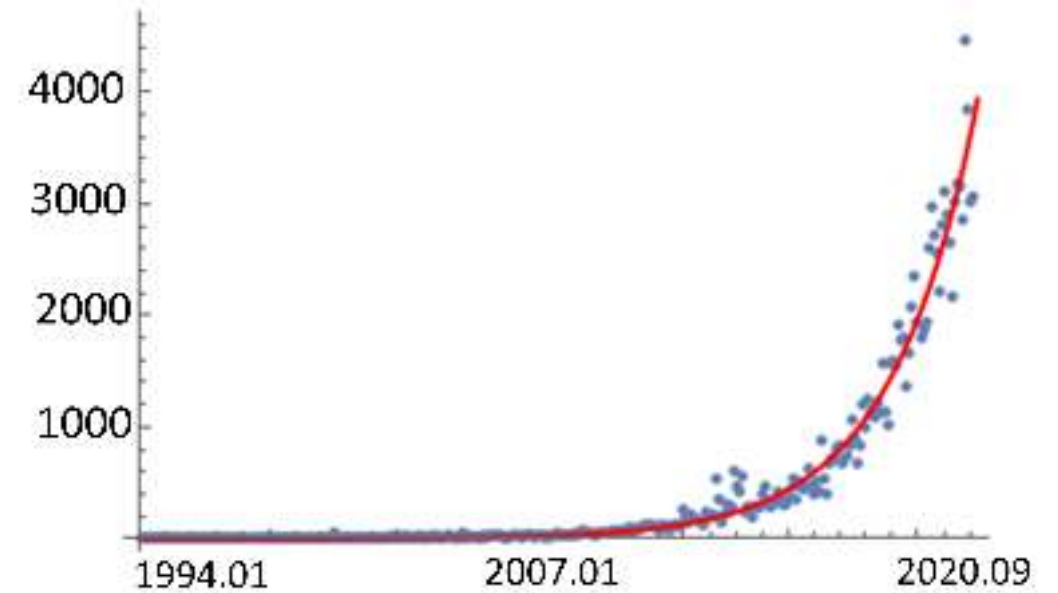


2012 to Present: Deep Learning Explosion

CVPR Papers

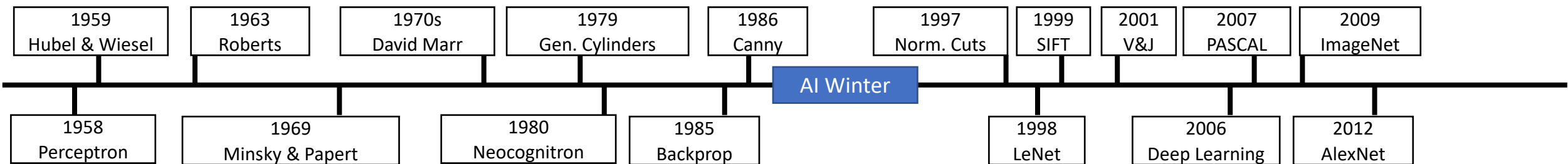


ML+AI arXiv papers per month



Publications at top Computer Vision conference

arXiv papers per month ([source](#))



2012 to Present: Deep Learning is Everywhere

Image Classification



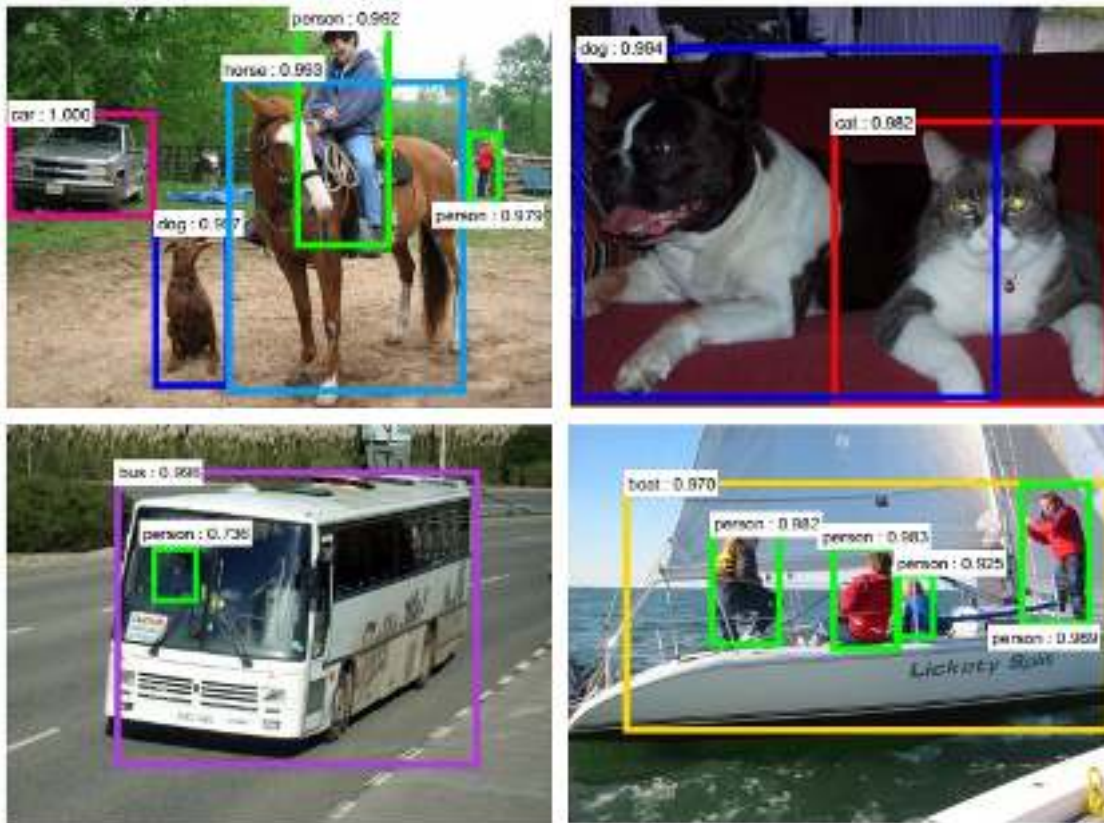
Image Retrieval



Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

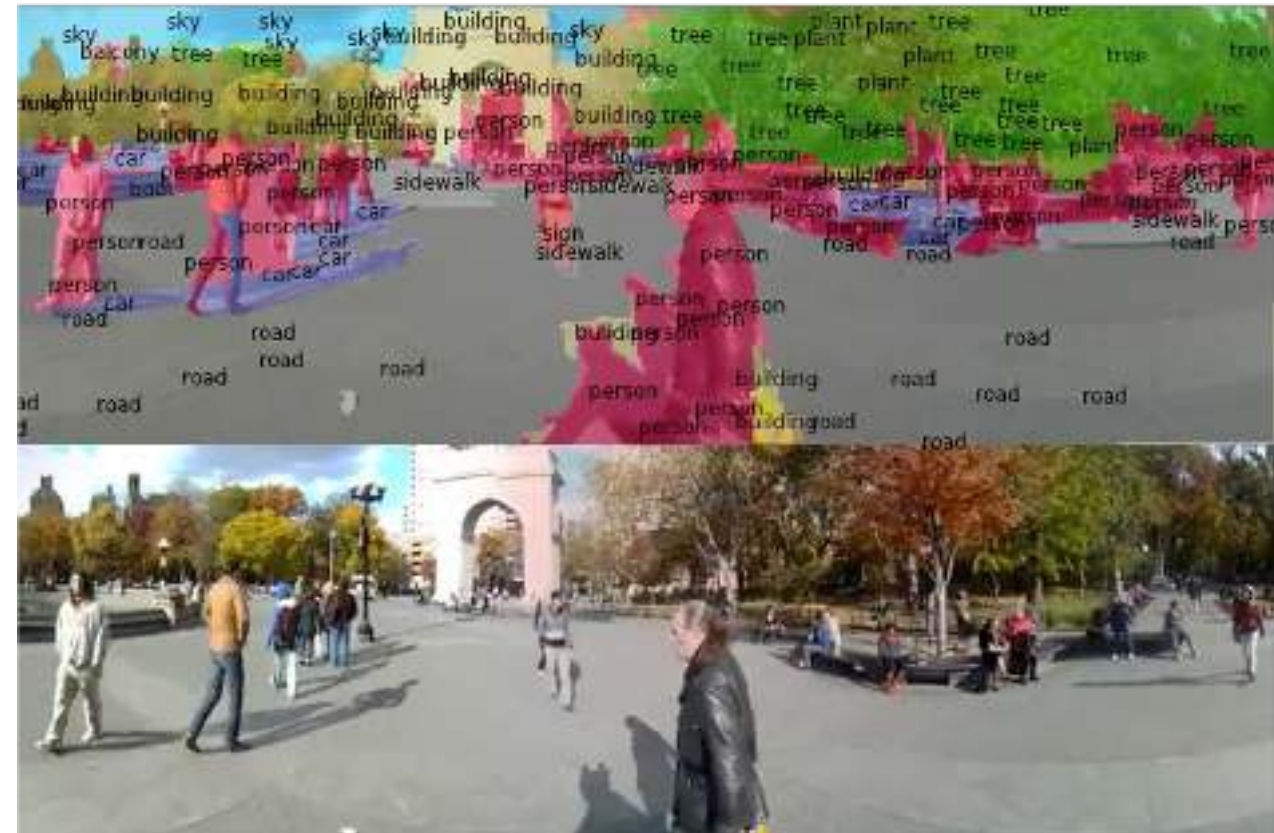
2012 to Present: Deep Learning is Everywhere

Object Detection



Ren, He, Girshick, and Sun, 2015

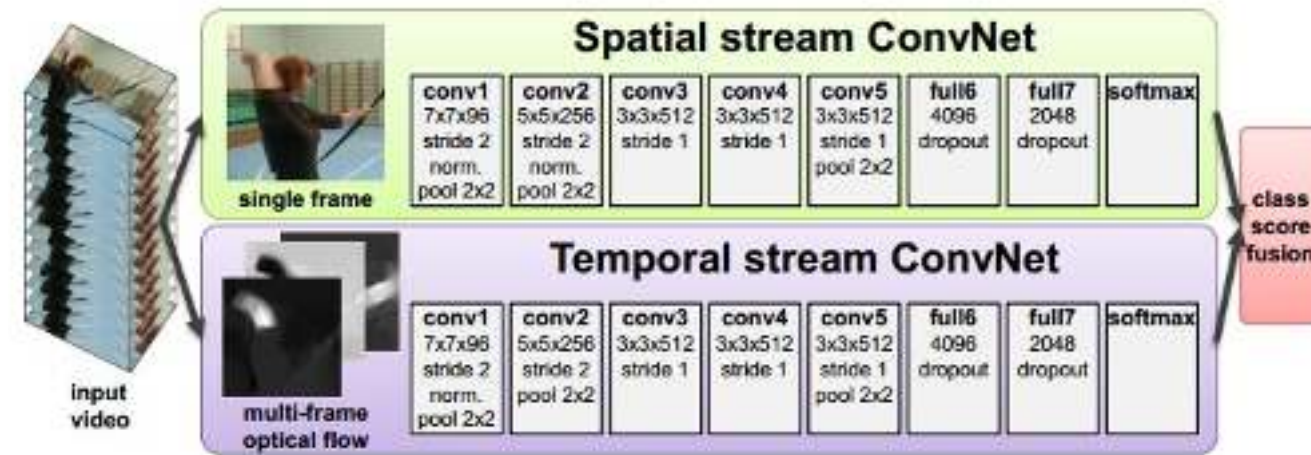
Image Segmentation



Fabaret et al, 2012

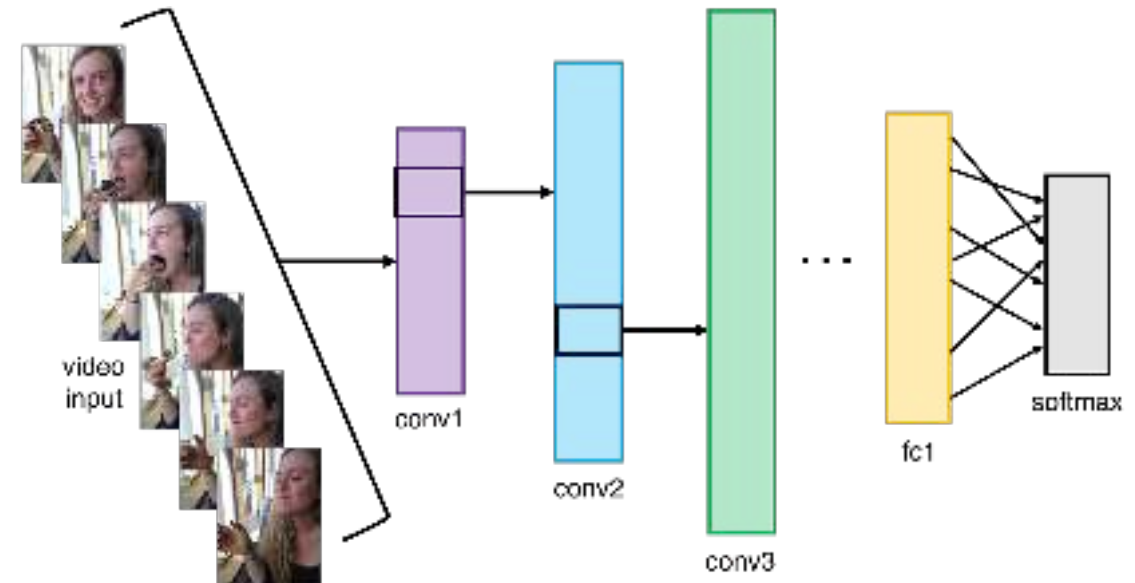
2012 to Present: Deep Learning is Everywhere

Video Classification



Simonyan et al, 2014

Activity Recognition

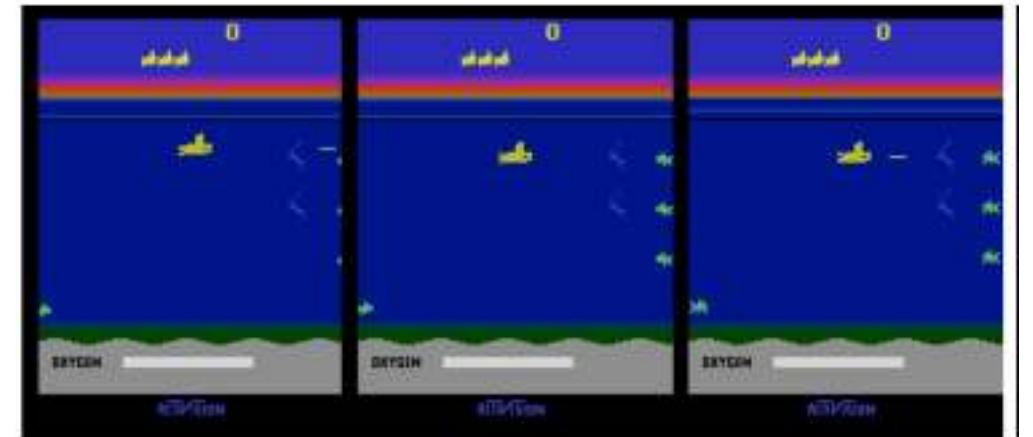
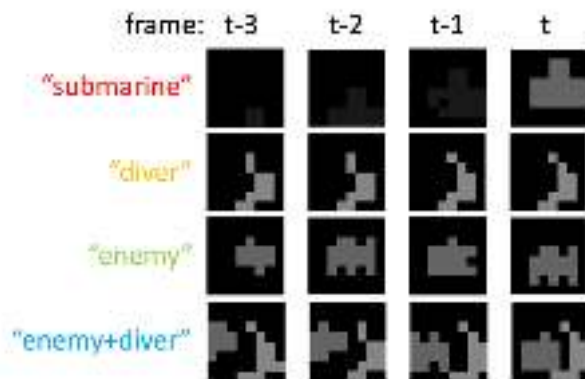


2012 to Present: Deep Learning is Everywhere

Pose Recognition (Toshev and Szegedy, 2014)

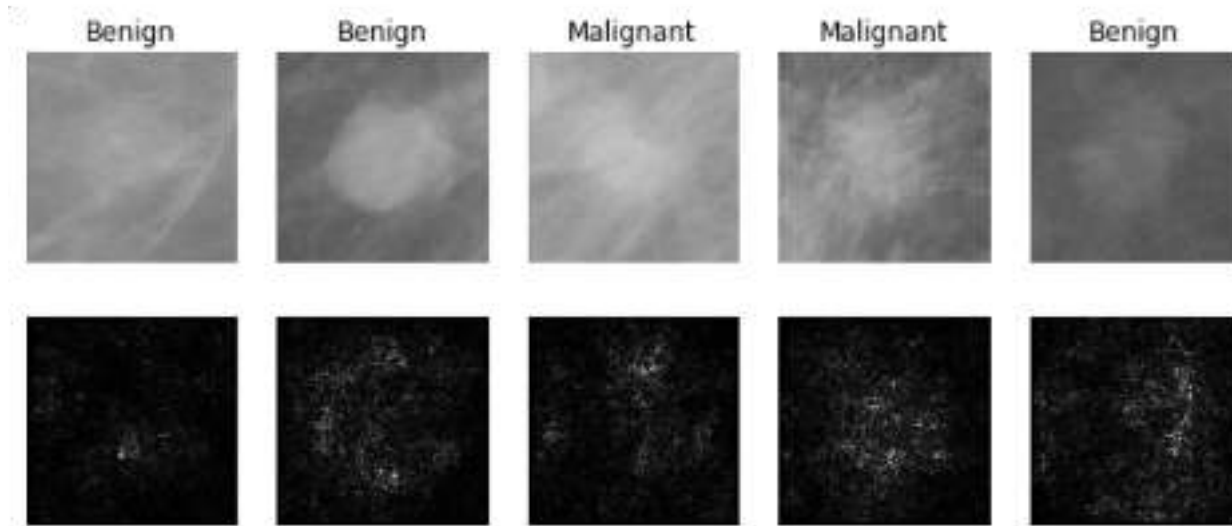


Playing Atari games (Guo et al, 2014)



2012 to Present: Deep Learning is Everywhere

Medical Imaging



Levy et al, 2016 Figure reproduced with permission

Galaxy Classification



Dieleman et al, 2014

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Whale recognition



[Kaggle Challenge](#)

This image by Christin Khan is in the public domain and originally came from the U.S. NOAA.

2012 to Present: Deep Learning is Everywhere



*A white teddy bear
sitting in the grass*



*A man in a baseball
uniform throwing a ball*



*A woman is holding
a cat in her hand*

Image Captioning

Vinyals et al, 2015

Karpathy and Fei-Fei, 2015



*A man riding a wave
on top of a surfboard*



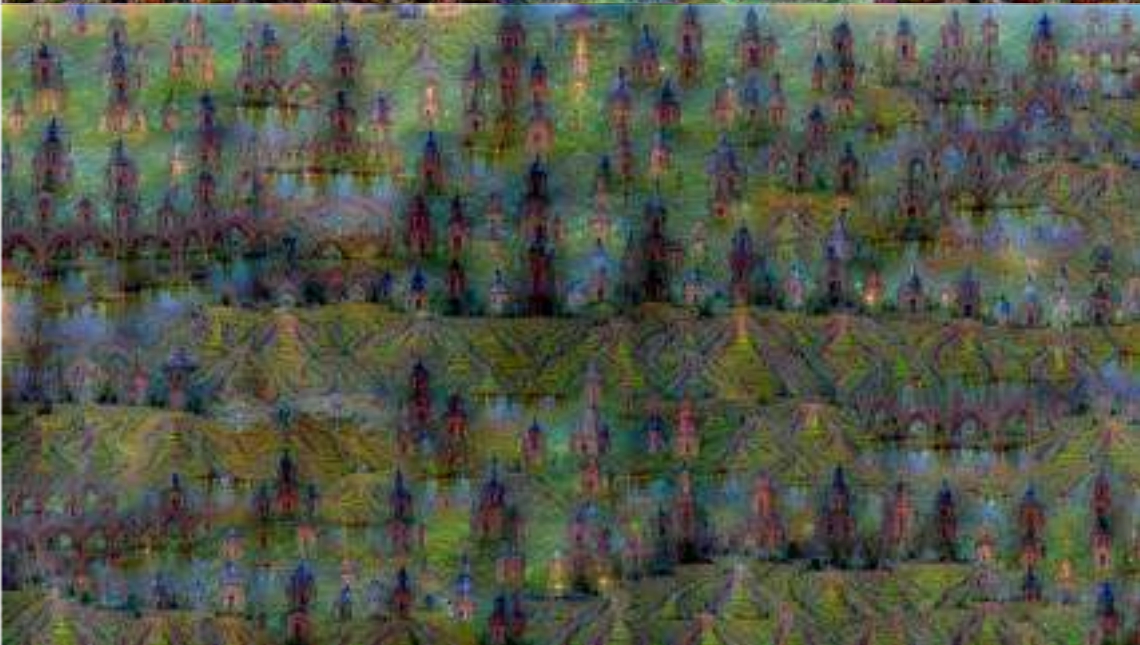
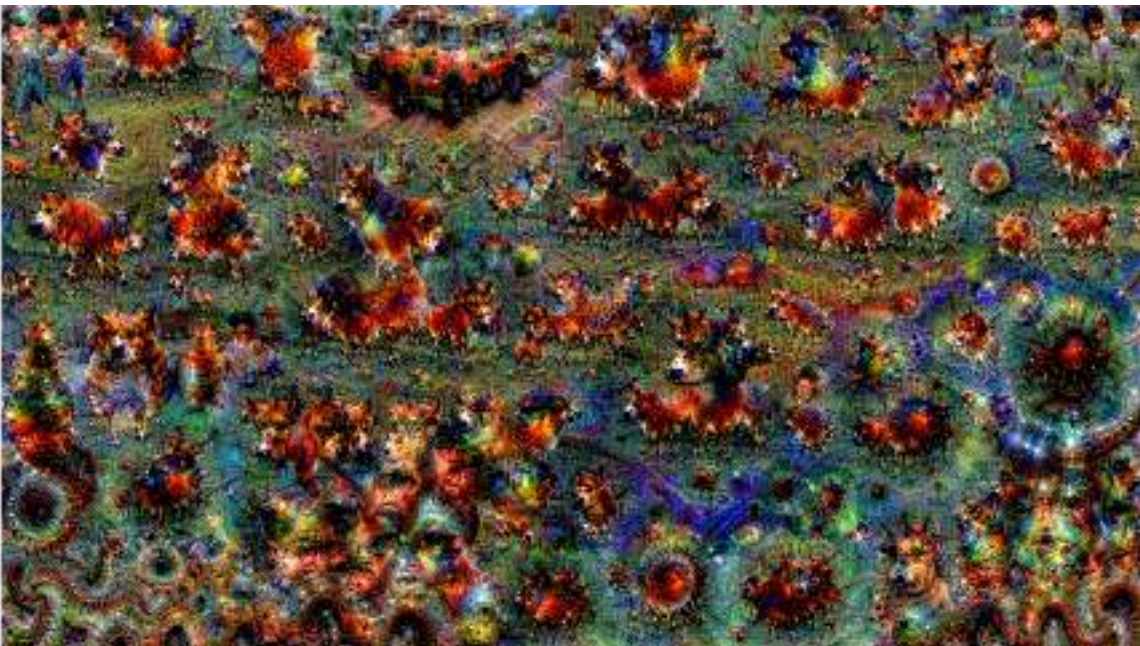
*A cat sitting on a
suitcase on the floor*



*A woman standing on a
beach holding a surfboard*

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Captions generated by Justin Johnson using [NeuralTalk2](#)

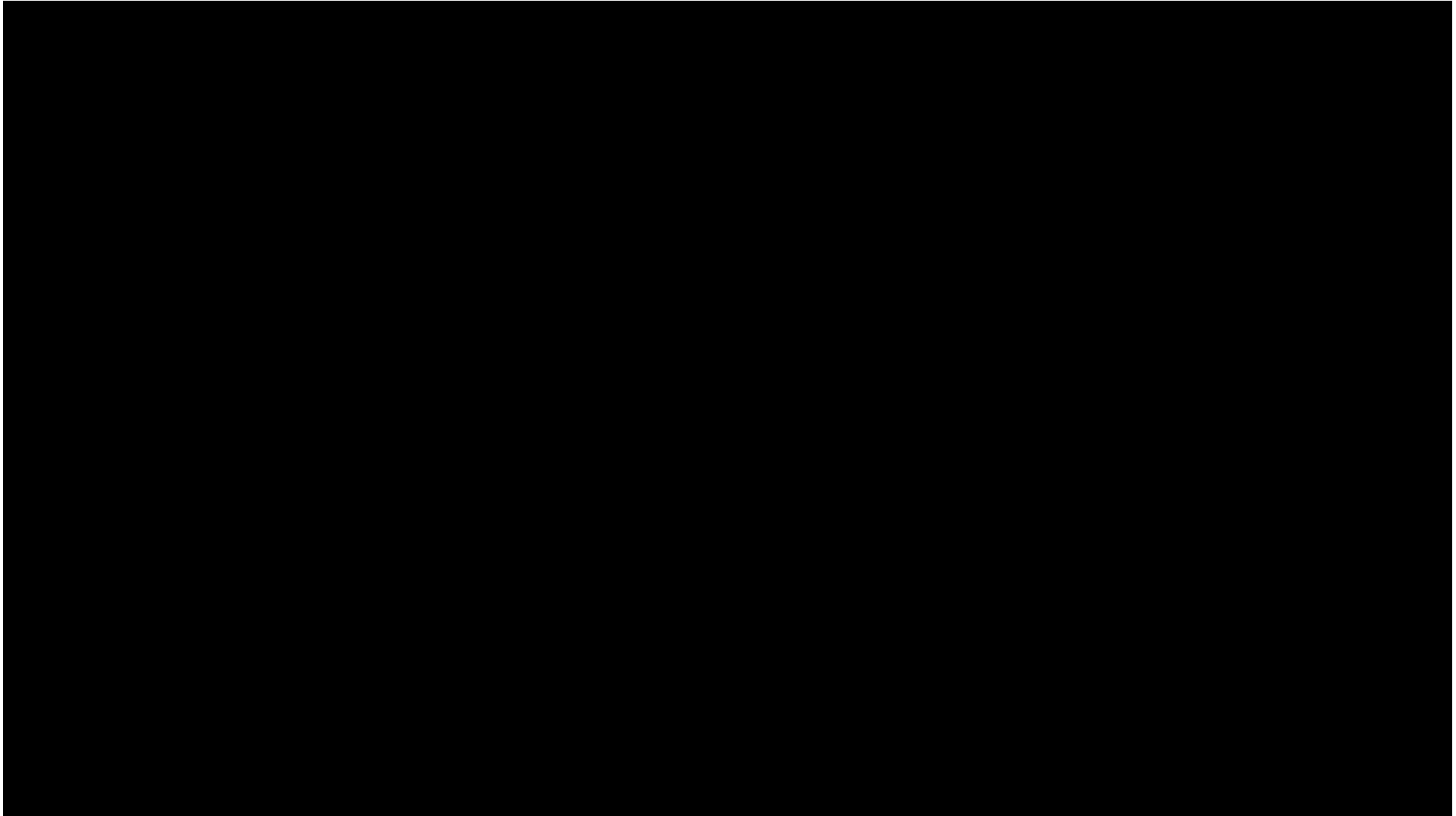


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Bokeh image is in the public domain
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Figures copyright Justin Johnson, 2015. Reproduced with permission. Generated using the Inceptionism approach from a [blog post](#) by Google Research.

Mordvinsev et al, 2015
 Gatys et al, 2016

2012 to Present: Deep Learning is Everywhere



Karras et al, "Progressive Growing of GANs for Improved Quality, Stability, and Variation", ICLR 2018

2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

an armchair in the shape of an avocado. an armchair imitating an avocado.

AI-GENERATED IMAGES



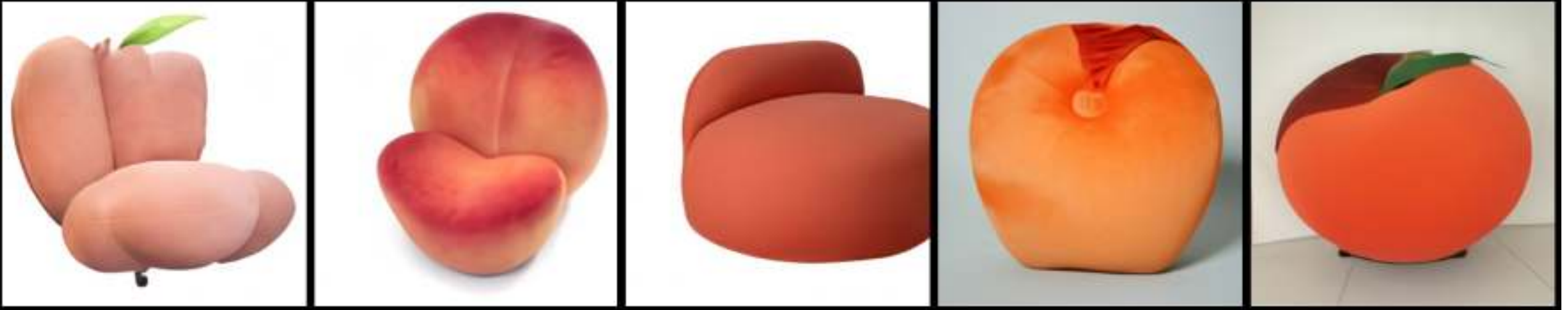
Ramesh et al, "DALL·E: Creating Images from Text", 2021. <https://openai.com/blog/dall-e/>

2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

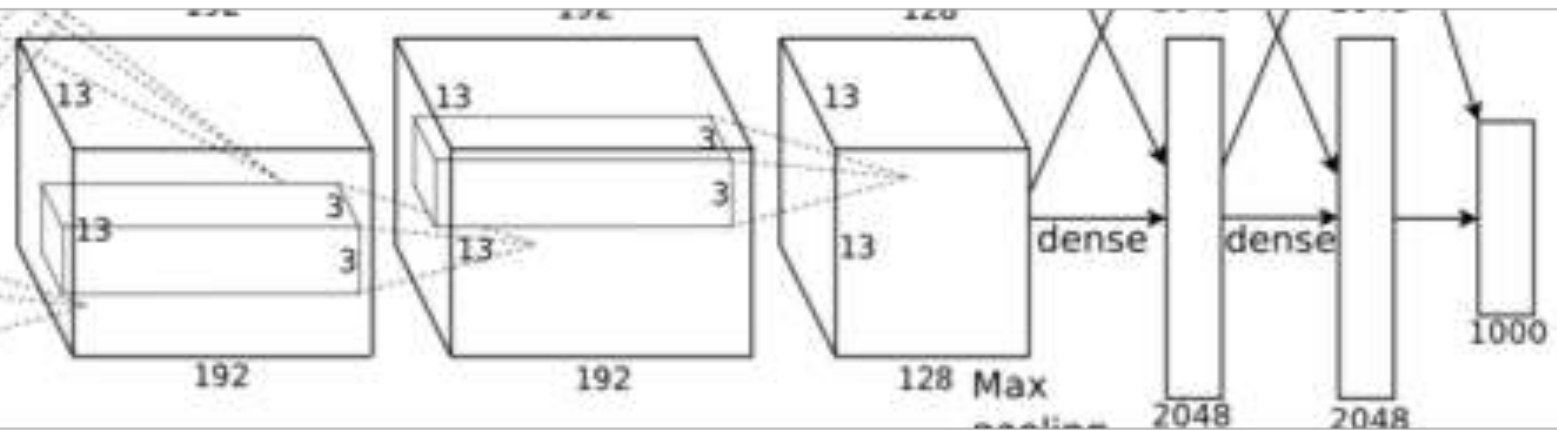
an armchair in the shape of a peach. an armchair imitating a peach.

AI-GENERATED IMAGES



Ramesh et al, "DALL·E: Creating Images from Text", 2021. <https://openai.com/blog/dall-e/>

Algorithms



Data

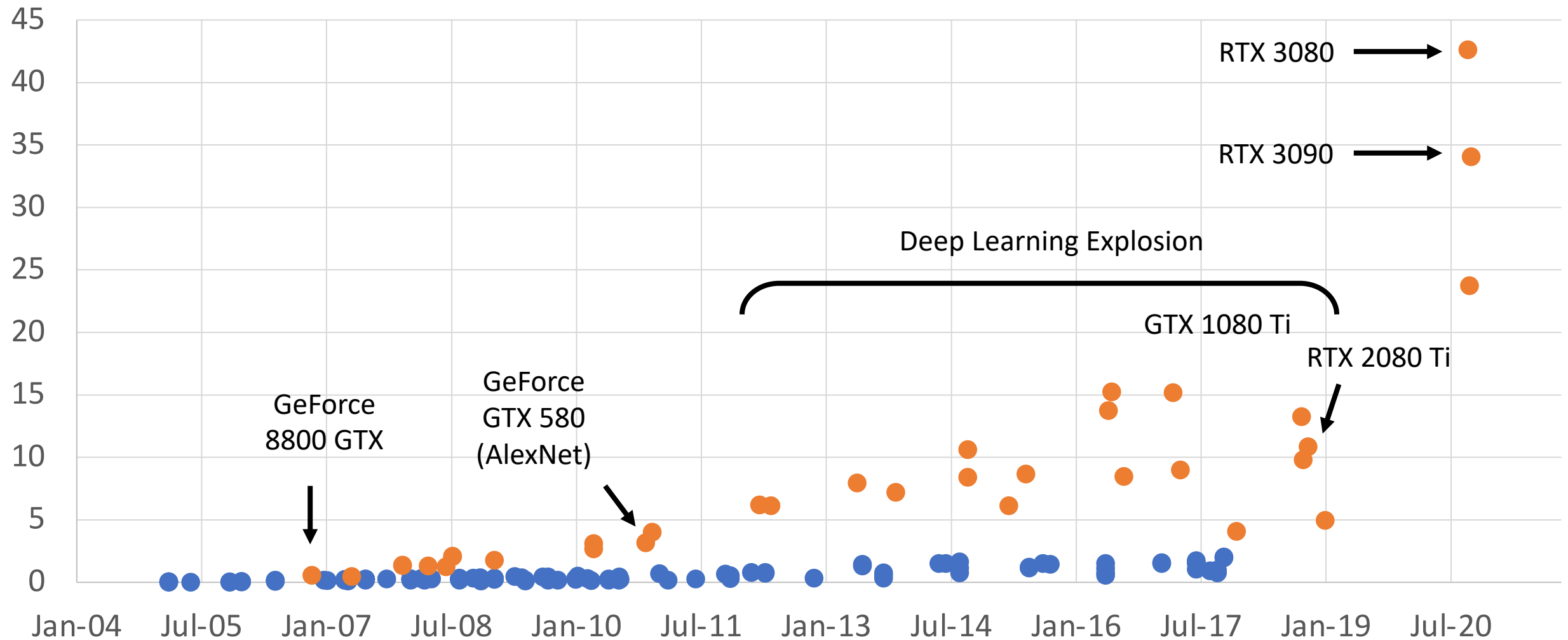


Computation



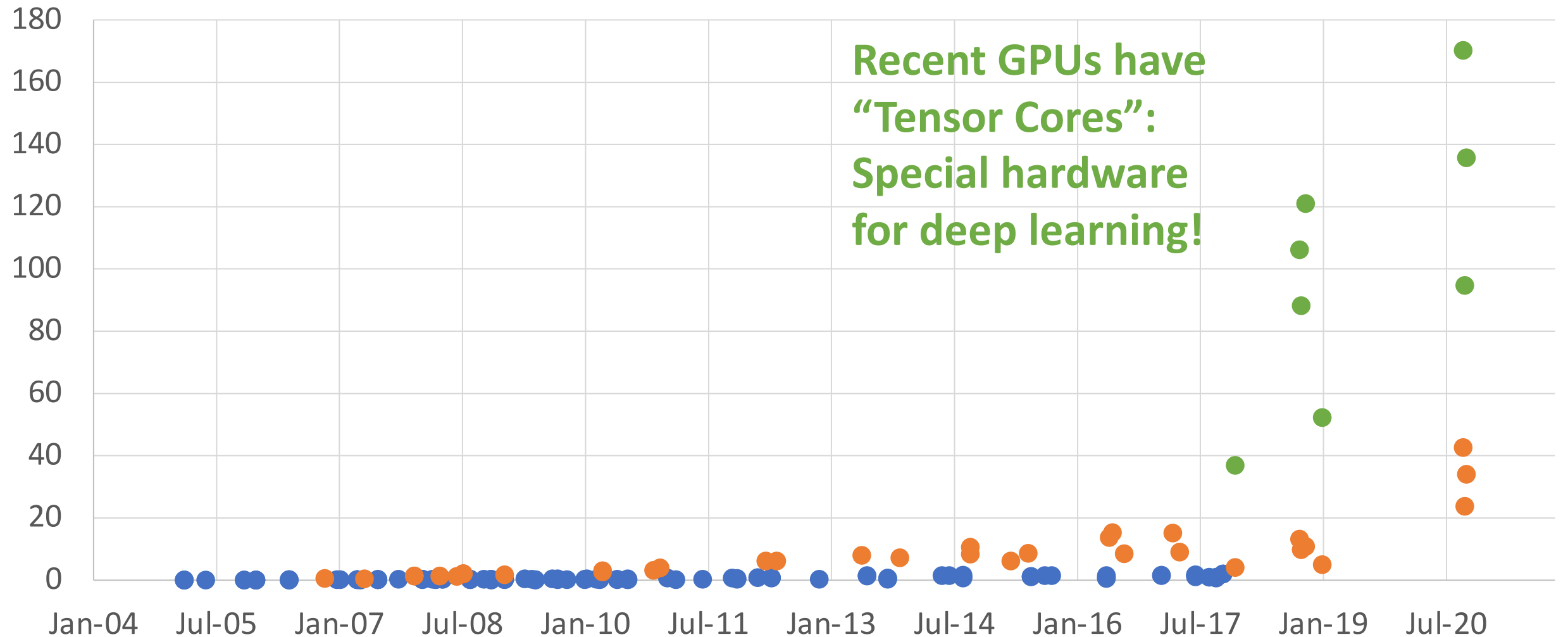
GFLOP per Dollar

● CPU ● GPU (FP32)



GFLOP per Dollar

● CPU ● GPU (FP32) ● GPU (Tensor Core)



2018 Turing Award



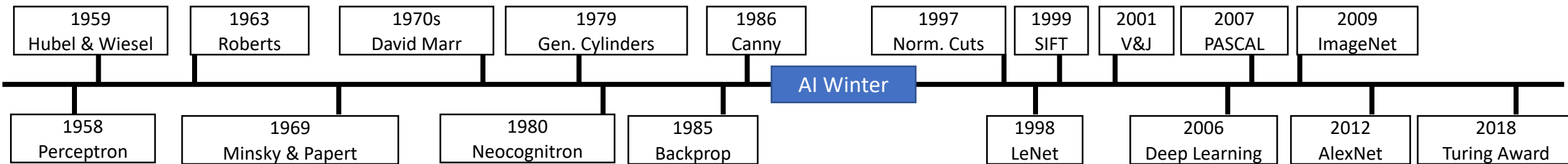
Yoshua Bengio



Geoffrey Hinton



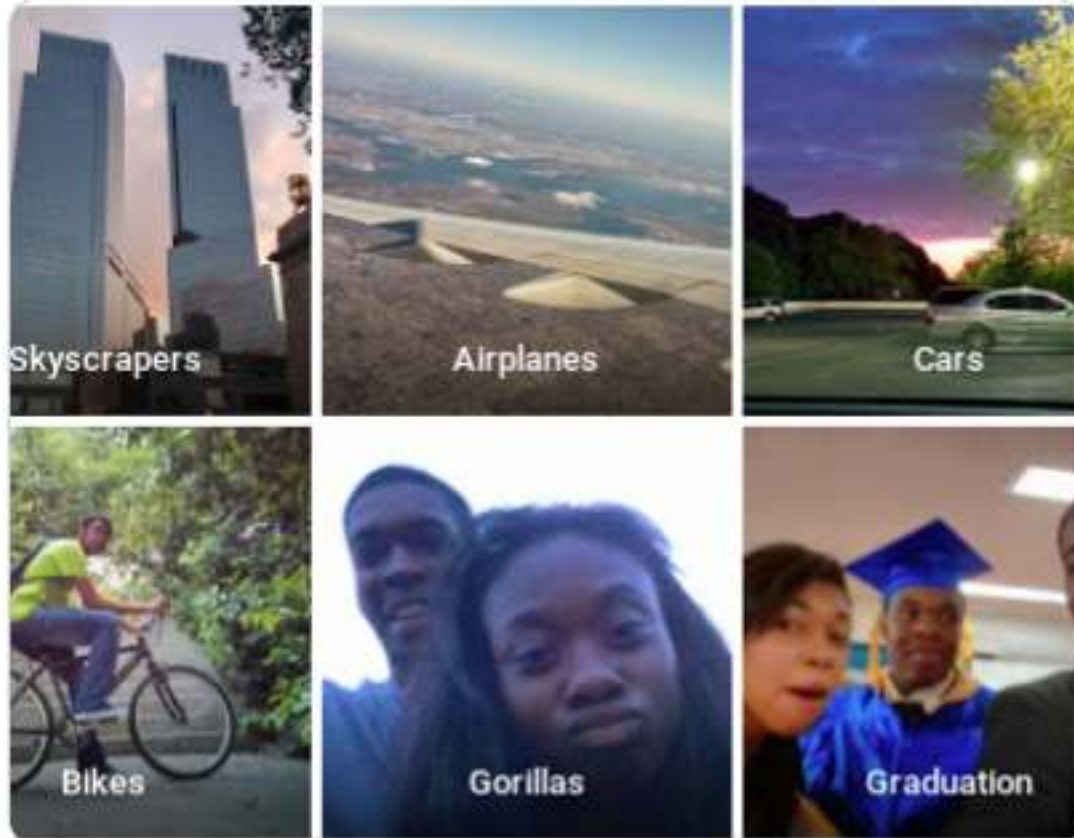
Yann LeCun



Despite our success, computer vision still has a long way to go...

Computer Vision can cause harm

Harmful Stereotypes



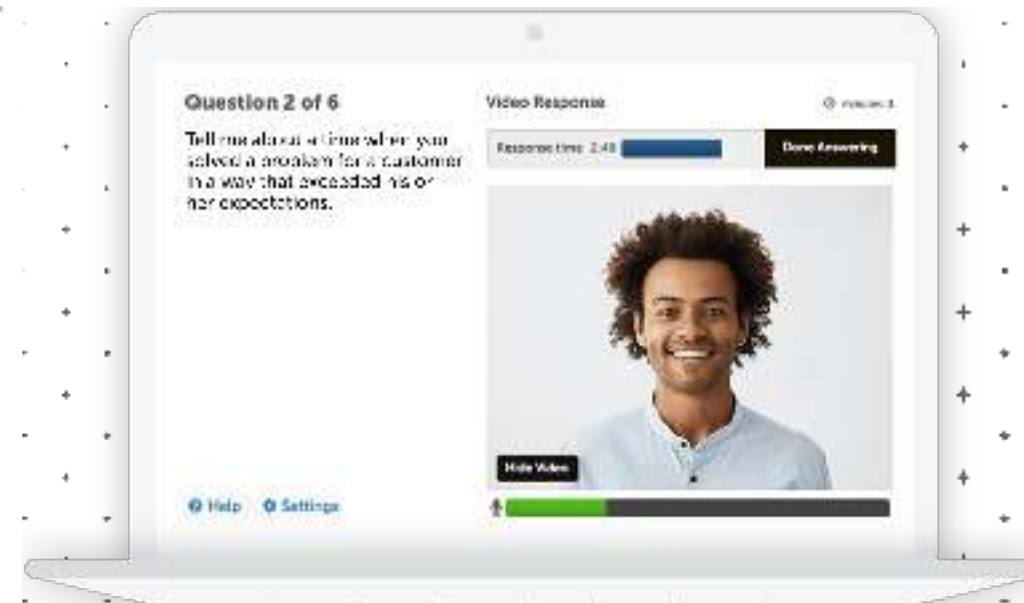
Barocas et al, "The Problem With Bias: Allocative Versus Representational Harms in Machine Learning", SIGCIS 2017
Kate Crawford, "The Trouble with Bias", NeurIPS 2017 Keynote
Source: <https://twitter.com/jackyalcine/status/615329515909156865> (2015)

Affect people's lives

Technology

A face-scanning algorithm increasingly decides whether you deserve the job

HireVue claims it uses artificial intelligence to decide who's best for a job. Outside experts call it 'profoundly disturbing.'



Source: <https://www.washingtonpost.com/technology/2019/10/22/ai-hiring-face-scanning-algorithm-increasingly-decides-whether-you-deserve-job/>
<https://www.hirevue.com/platform/online-video-interviewing-software>

Example Credit: Timnit Gebru

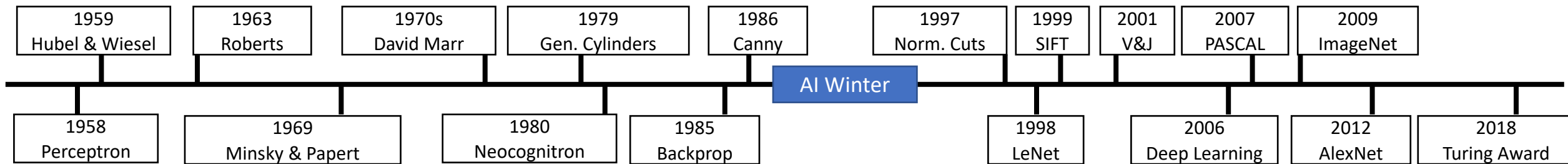


This image is copyright-free United States government work

Example credit: Andrej Karpathy

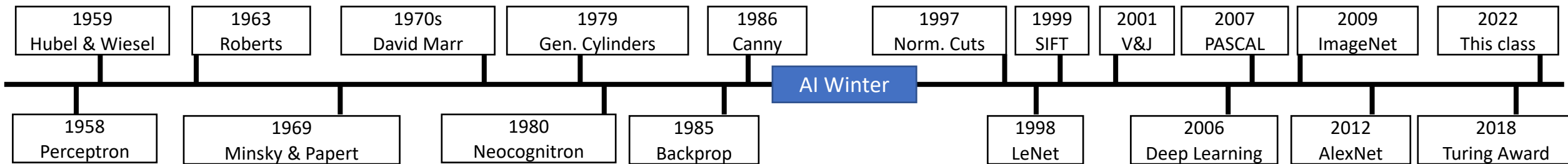
Today's Agenda

- A brief history of computer vision and deep learning
- Course overview and logistics



Today's Agenda

- A brief history of computer vision and deep learning
- Course overview and logistics



Course Staff

Instructor



Justin Johnson
Assistant Professor, CSE

GSI / IAs



Karan Desai (KD)



Janpreet Singh (JS)



Jim Yang



Wallace Sui (WS)



Gaurav Kaul

How to contact us

- Course Website: <https://web.eecs.umich.edu/~justincj/teaching/eecs498/>
 - Syllabus, schedule, assignments, slides, lecture videos, etc
- Piazza: <https://piazza.com/class/kxtai72amx34p0>
 - (Almost) all questions about the course should go here!
 - We will also use Piazza to communicate with you
 - Use private questions if you want to post code
- EECS Autograder: <https://autograder.io/web/course/151>
 - For turning in homework assignments
- [Google Calendar](#): For office hours (starting next week)
- Email: Only for sensitive, confidential issues

Course Website: Check the Schedule!

 EECS 498.008 / 598.008 Deep Learning for Computer Vision Winter 2022			
Schedule			
Lectures are Mondays and Wednesdays, 4:30pm to 6pm. Attendance is not required. Recordings will be posted after each lecture in case you are unable to attend the scheduled time.			
Some lectures have reading drawn from the course notes of Stanford CS 231n, written by Andrej Karpathy .			
Some lectures have optional reading from the book <i>Deep Learning</i> by Ian Goodfellow, Yoshua Bengio, and Aaron Courville (DLC for short). The entire text of the book is available for free online so you don't need to buy a copy.			
Event	Date	Description	Course Materials
Lecture 1	Wednesday January 5	Course introduction Computer vision overview Historical context Course logistics	[slides] [FA2019 video (public)] [Python tutorial] [DLC Sec 1.2] [DLC Sec 6.6]
Lecture 2	Monday January 10	Image Classification Data-driven approach K-Nearest Neighbor Hyperparameters Cross-validation	[slides] [FA2019 video (public)] [231n Image Classification]
Lecture 3	Wednesday January 12	Linear Classifiers Algebraic / Visual / Geometric viewpoints Softmax / SVM classifiers	[slides] [FA2019 video (public)] [231n Linear Classification]
A1 Due	Friday January 14	Assignment 1 Due PyTorch warmup iNN Classifier	[Assignment 1]
	Monday January 17	No class MLK Day	

<https://web.eecs.umich.edu/~justincj/teaching/eecs498/WI2022/schedule.html>

Piazza Etiquette

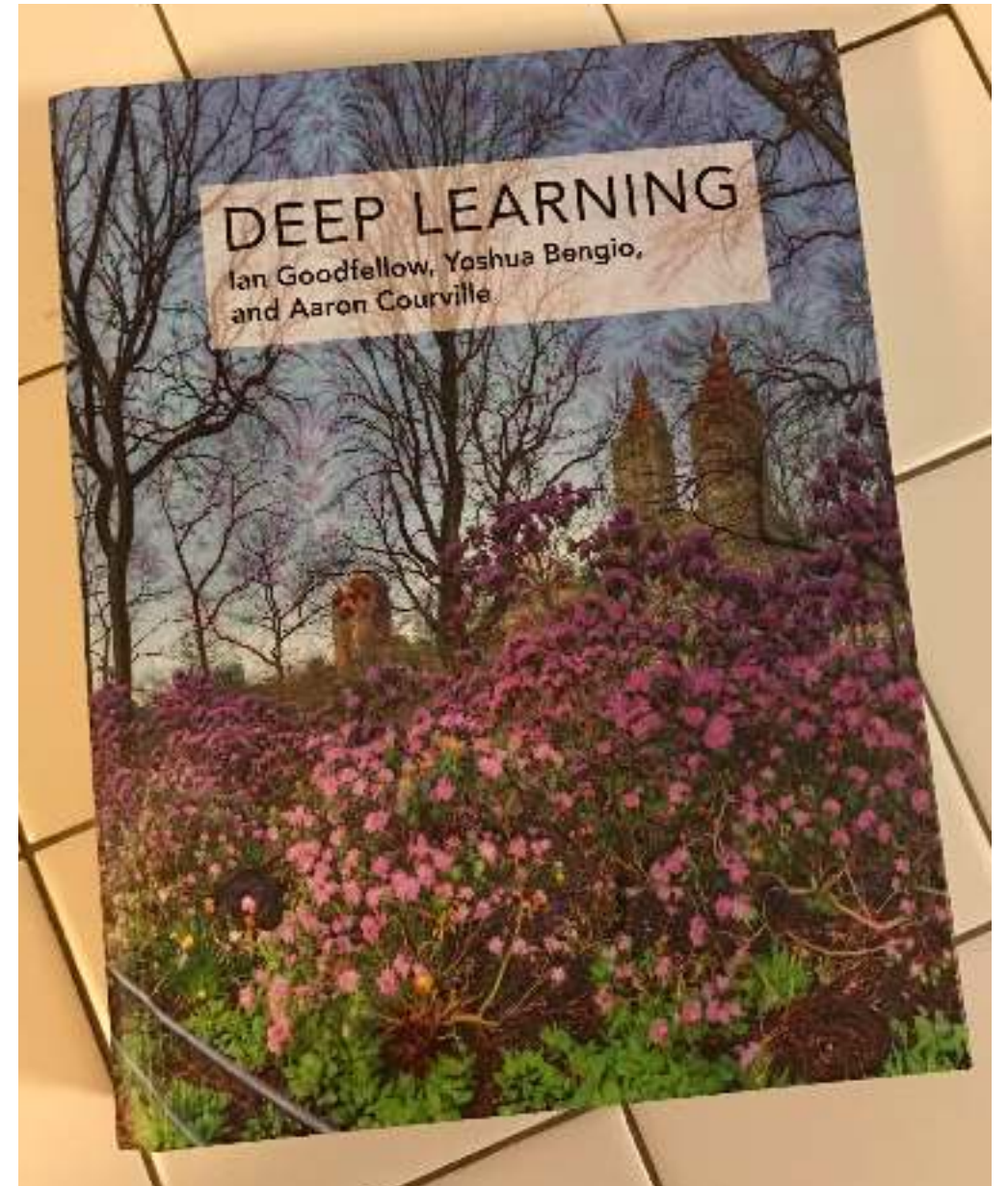
- Post only short snippets of code (< 20-30 lines)
- Ask a specific, concrete question
- Explain what you have tried so far, and what happened
- See StackOverflow guide on asking good questions:
<https://stackoverflow.com/help/how-to-ask>

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- Don't expect an answer within 30 minutes of posting
- Monday – Friday, 10am – 6pm EST we'll try to answer within 2 hours
- Other times, we'll try to answer within 12 hours

Optional Textbook

- [Deep Learning](#) by Goodfellow, Bengio, and Courville
- [Free online](#)



Course Content and Grading

- 6 programming assignments (10% each)
- Midterm Exam (20%)
- Mini-Project (20%)
- Late policy
 - 3 free late days to use on assignments
 - Once free late days are exhausted, 25% penalty per day

Programming Assignments

- Python, PyTorch, will use Google Colab
- “Earn your wings” – implement things “from scratch” in early assignments, then use PyTorch in later assignments
- “Challenge Questions”
 - Go above and beyond the basic expectations of each assignment
 - Much higher time/points ratio than other parts of the assignment
 - Not necessary to get an A: will be 5% or less of each assignment

Midterm Exam

- Written exam testing basic concepts from first half of course
- True / False, Multiple choice, short answer
- We will provide a practice exam to get a sense of the format of the questions (but not necessarily the length or difficulty)

Mini-Project

- New this year!
- Work in teams of 1 to 3
- Homework assignments: We provide starter code, you “fill in the blank”
- Mini-Project: We provide high-level written description of an algorithm to implement, you do the whole thing “from scratch”
- We will give ~3 project descriptions, you pick one of those
- Deliverable: zip of code, and Colab notebook that walks through your implementation and main results (should be like notebooks from homework!)

Collaboration Policy

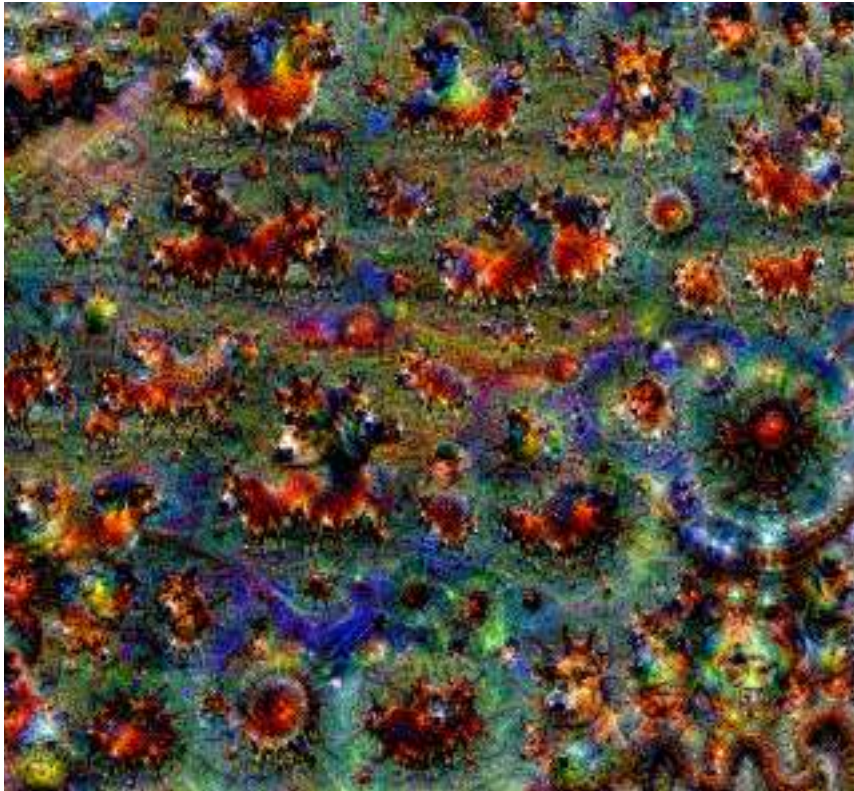
- **Rule 1:** Don't look at solutions or code that are not your own; everything you submit should be your own work
- **Rule 2:** Don't share your solution code with others; however discussing ideas or general strategies is fine and encouraged
- **Rule 3:** Indicate in your submissions anyone you worked with
- Turning in something late / incomplete is better than violating the honor code

Course Philosophy

- Thorough and Detailed.
 - This not “Learn PyTorch in 90 days”, nor “Deep Learning in 10 lines of code”
 - Understand how to write from scratch, debug, and train convolutional and other types of deep neural networks
 - We prefer to write from scratch, rather than rely on existing implementations
- Practical
 - Focus on practical techniques for training and debugging neural networks
 - Will use state-of-the-art software tools like PyTorch and TensorFlow
- State of the art
 - Most material we cover is research published in the last 5 years

Course Philosophy

- Will also cover some fun topics:
 - Image captioning
 - DeepDream, Artistic Style Transfer



Course Structure

- First half: Fundamentals
 - Details of how to implement and train different types of networks
 - Fully-connected networks, convolutional networks, recurrent networks
 - How to train and debug, very detailed
- Second half: Applications and “Researchy” topics
 - Object detection, image segmentation, 3D vision, videos
 - Attention, Transformers
 - Vision and Language
 - Generative models: GANs, VAEs, etc
 - Less detailed: provide overview and references, but skip some details

New Topics since FA2020

- Modern CNN architectures
 - SENets, MobileNets, NAS
 - EfficientNets, RegNets, NFNets
- Vision Transformers
 - Architectures: ViT, DeiT, Swin, MViT
 - Applications: DETR
 - MLP-like architectures
- Vision + Language
 - Language-based pretraining: CLIP, ALIGN
- Self-Supervised Learning
 - Contrastive learning
 - Masked autoencoding

First homework assignment

- Will be released by today or tomorrow
- Due Friday 1/14/2022
- Next lecture will be enough to complete it

Next time: Image Classification